



Dimensioning



TOPICS

- Introduction
- Dimensioning components
- Dimensioning object' s features
- Placement of dimensions.

Introduction



ENGINEERING DESIGN

PROCESS

RESULT

TRANSFERRED
INFORMATION

**Design
a part**



**Create
drawings**



Manufacture

**Sketches
of ideas**

**Multiview
Drawing**

Dimensioning

Shape

1. Size, Location
2. Non-graphic information

DEFINITION

Dimensioning is the process of specifying part's information by using of **figures**, **symbols** and **notes**.

This information are such as:

1. **Sizes and locations of features**
2. **Material's type**
3. **Number required**
4. **Kind of surface finish**
5. **Manufacturing process**
6. **Size and geometric tolerances**

This course

DIMENSIONING SYSTEM

1. Metric system : ISO standards

Examples 32, 32.5, 32.55, 0.5 (*not .5*) etc.

2. Decimal-inch system

Examples 0.25 (*not .25*), 5.375 etc.

3. Fractional-inch system

Examples $\frac{1}{4}$, $5\frac{3}{8}$ etc.

Dimensioning Components



DIMENSIONING COMPONENTS

- Extension lines
- Dimension lines
(*with arrowheads*)
- Leader lines

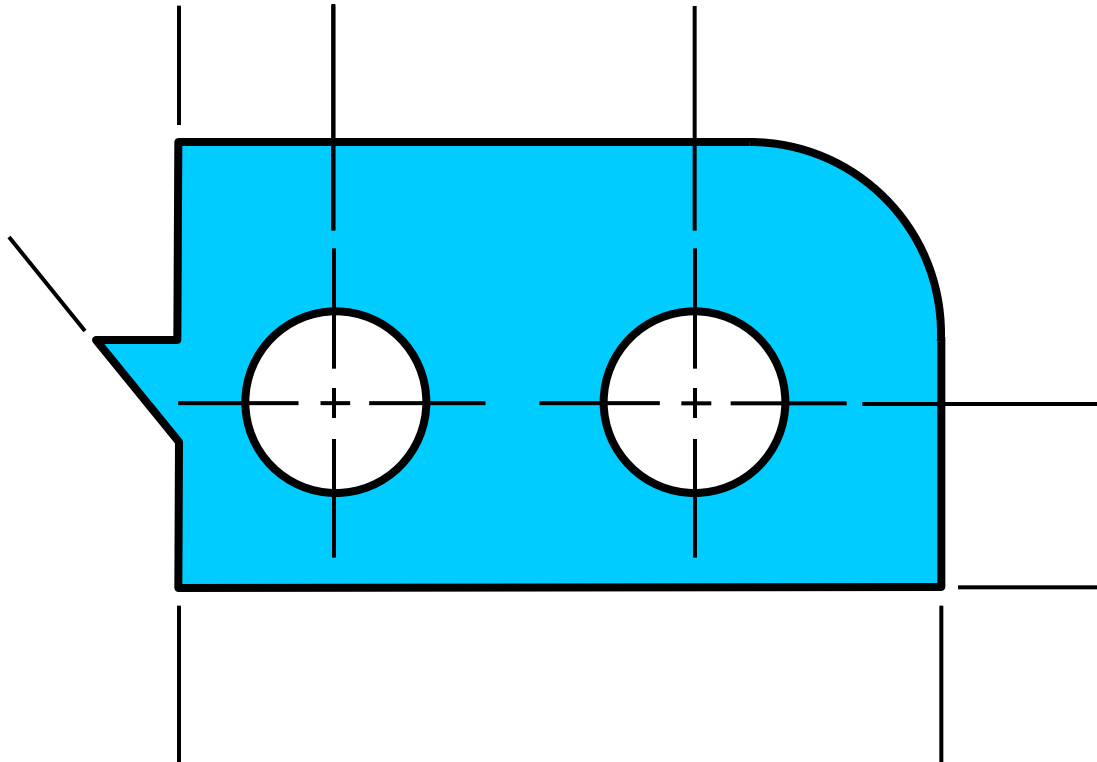
Drawn with
4H pencil

- Dimension figures
- Notes :
 - *local note*
 - *general note*

Lettered with
2H pencil.

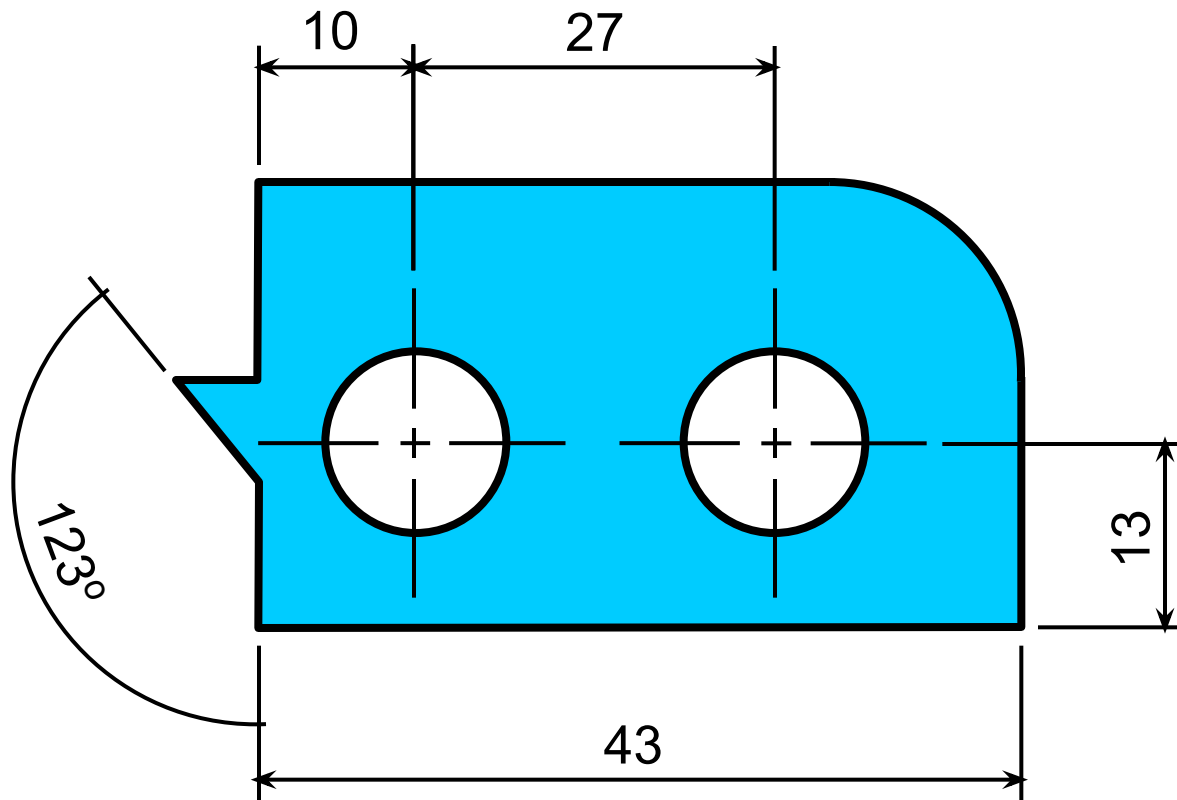
EXTENSION LINES

indicate the location on the object's features that are dimensioned.



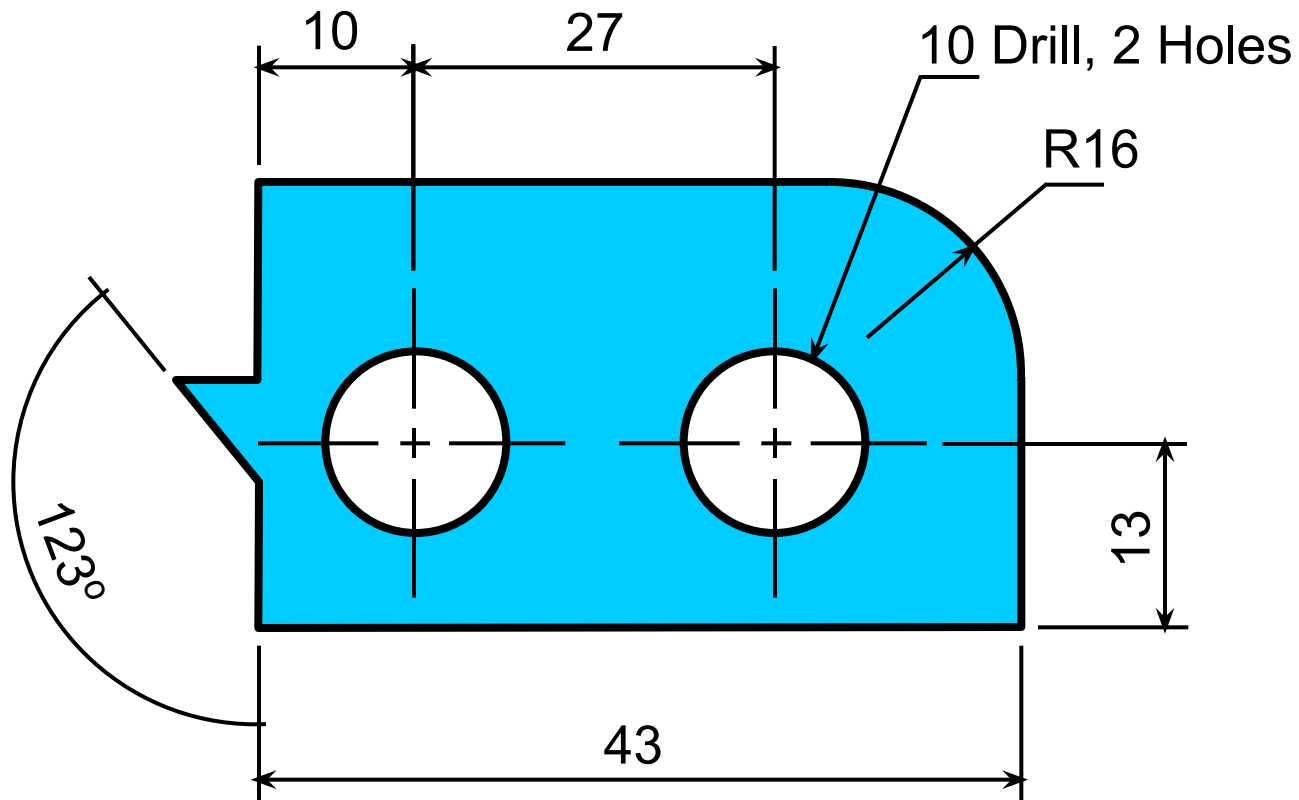
DIMENSION LINES

indicate the direction and extent of a dimension, and inscribe *dimension figures*.



LEADER LINES

indicate details of the feature with a *local* note.

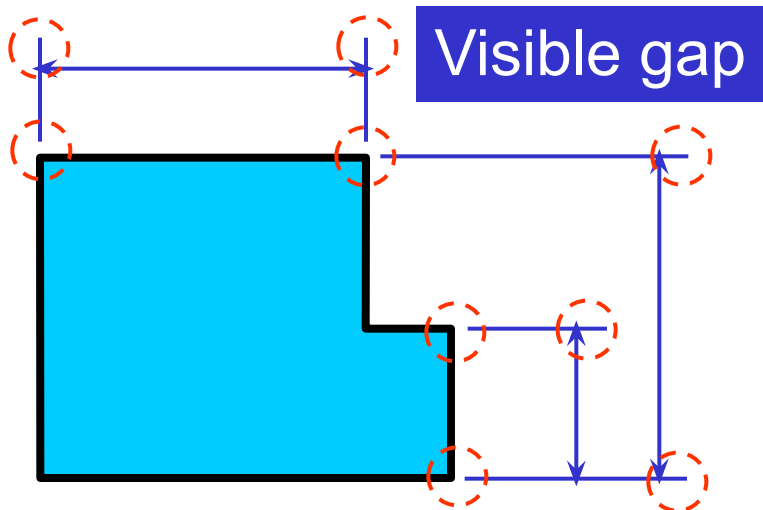


Recommended Practices

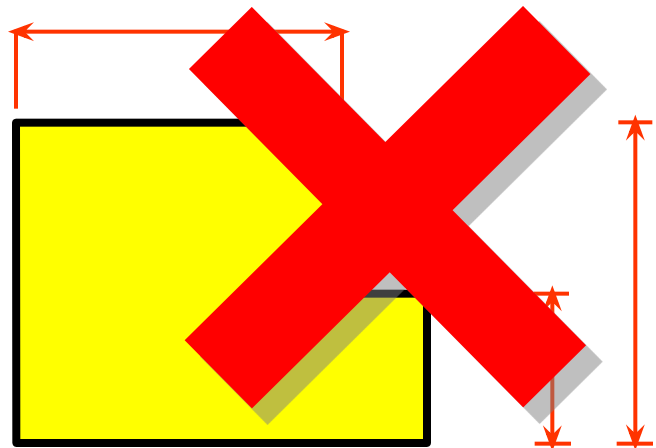


EXTENSION LINES

- Leave a **visible gap** (≈ 1 mm) from a view and start drawing an extension line.
- Extend the lines beyond the (last) dimension line 1-2 mm.

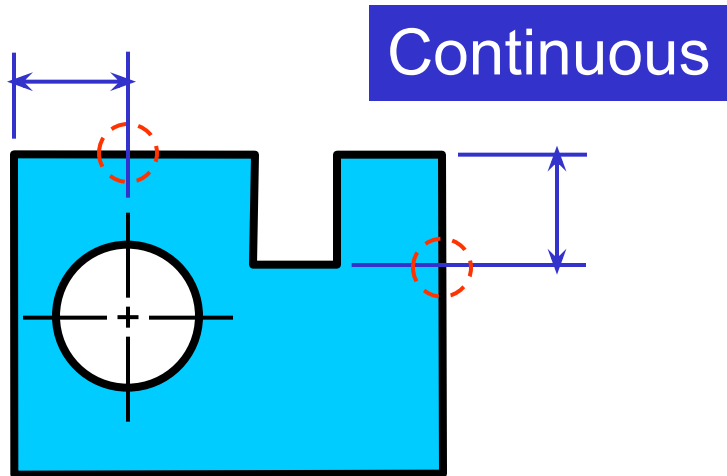


COMMON MISTAKE

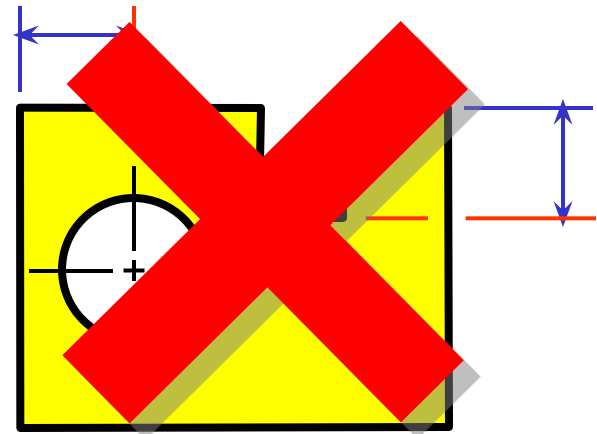


EXTENSION LINES

- **Do not** break the lines as they cross object lines.

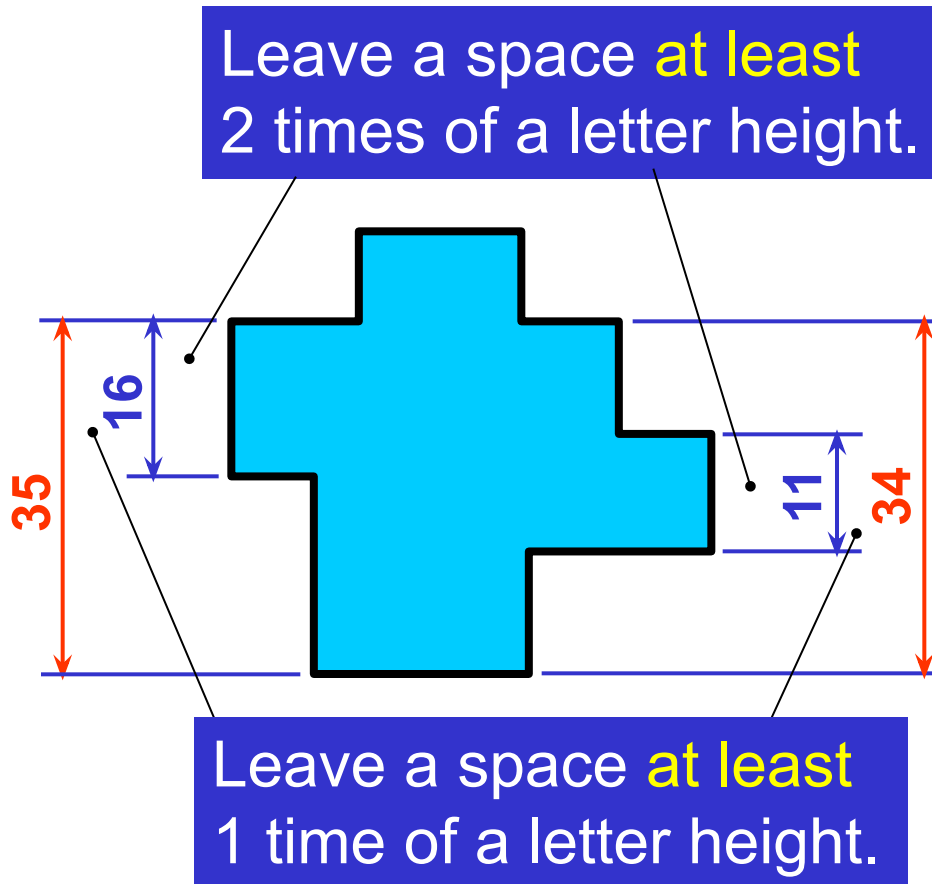


COMMON MISTAKE



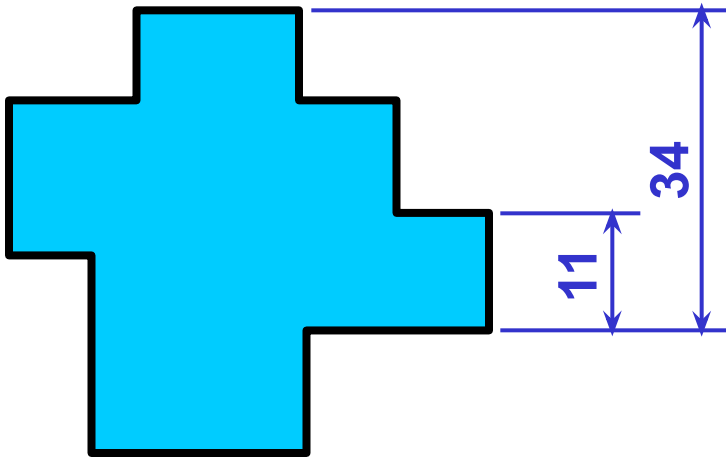
DIMENSION LINES

- Dimension lines should **not** be spaced too close to each other and to the view.

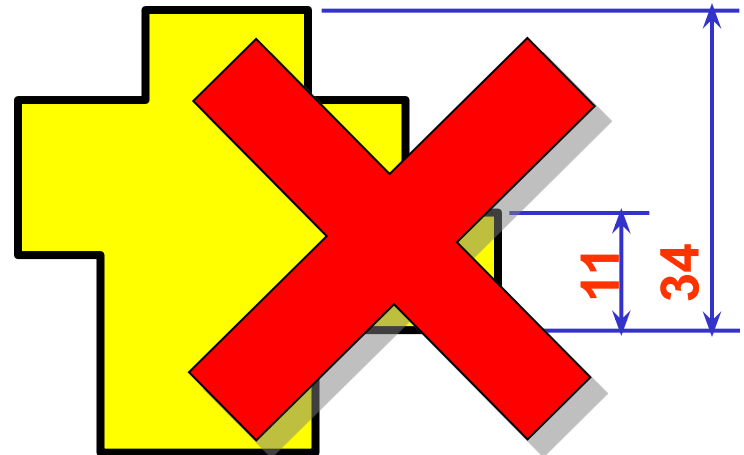


DIMENSION FIGURES

- Place the numbers at about 1 mm *above dimension line* and *between extension lines*.



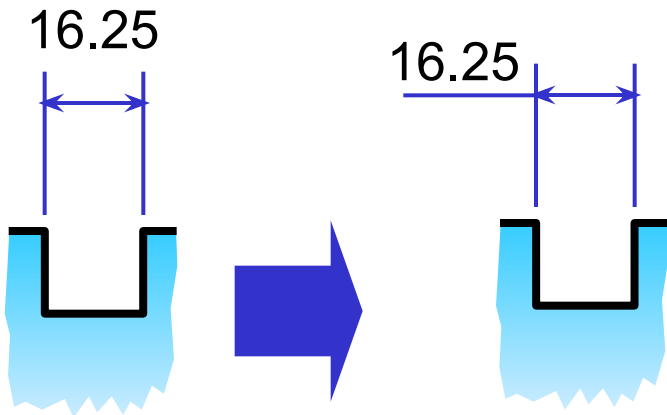
COMMON MISTAKE



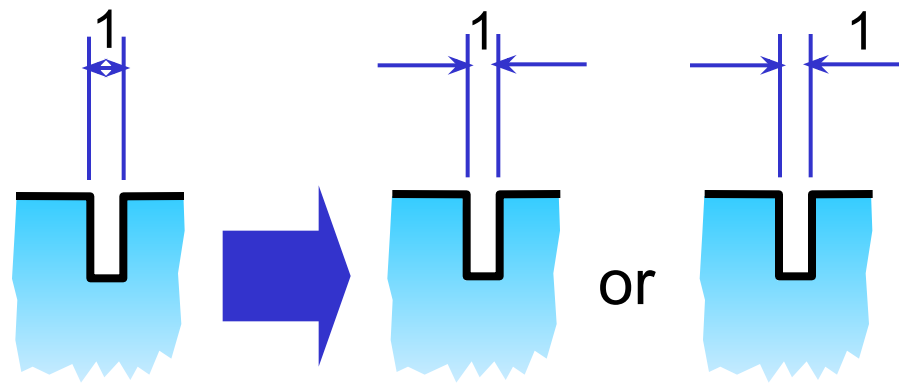
DIMENSION FIGURES

- When there is **not** enough space for figure or arrows, put it **outside** either of the extension lines.

Not enough space
for *figures*



Not enough space
for *arrows*



DIMENSION FIGURES : UNITS

The JIS and ISO standards adopt the unit of

- **Length** dimension in **millimeters** **without** specifying a unit symbol “mm”.
- **Angular** dimension in **degree** with a symbol “°” place behind the figures (and if necessary **minutes** and **seconds** may be used together).

DIMENSION FIGURES : ORIENTATION

1. Aligned method

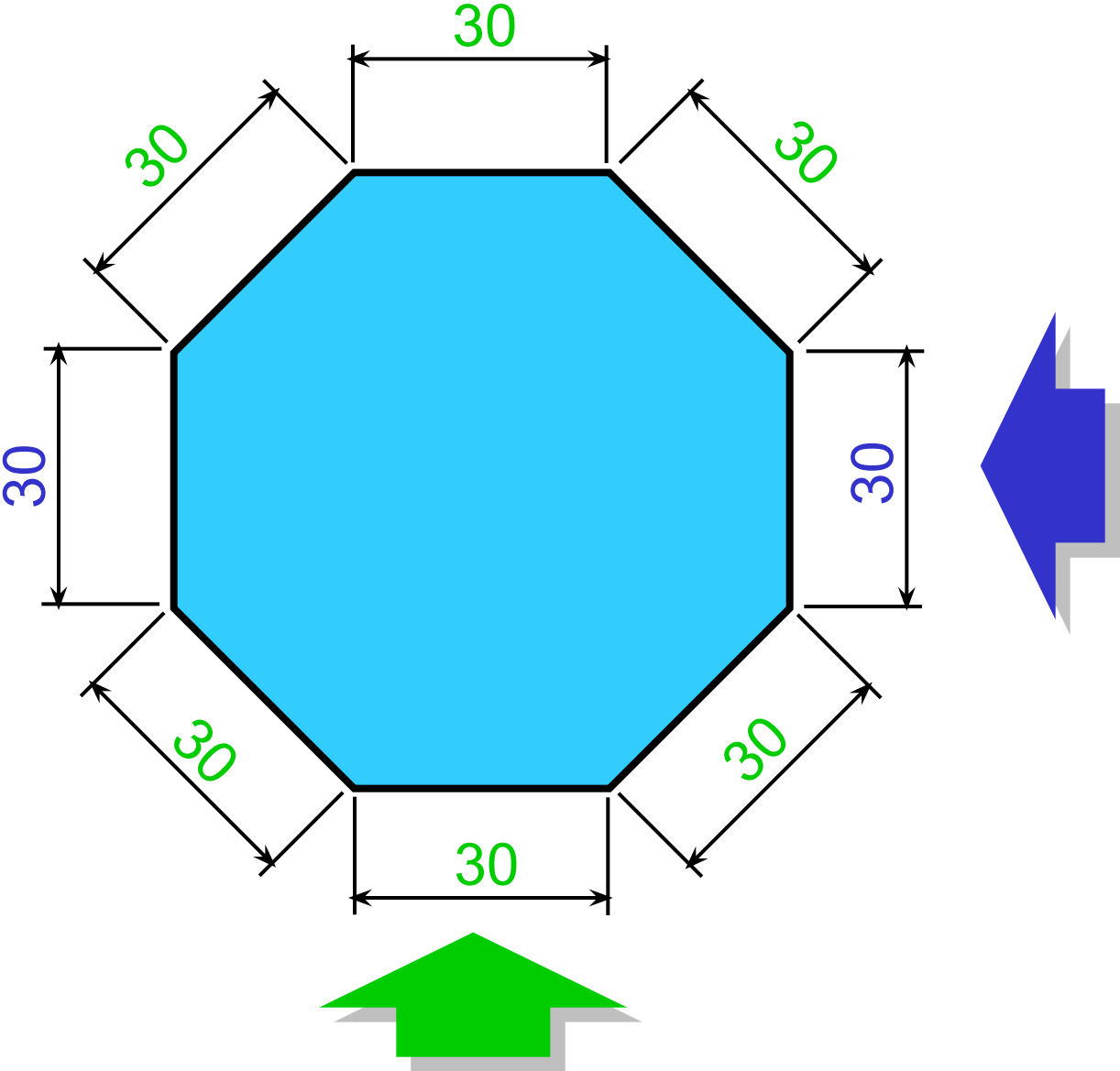
The dimension figures are placed so that they are readable from the **bottom** and **right side** of the drawing.

2. Unidirectional method

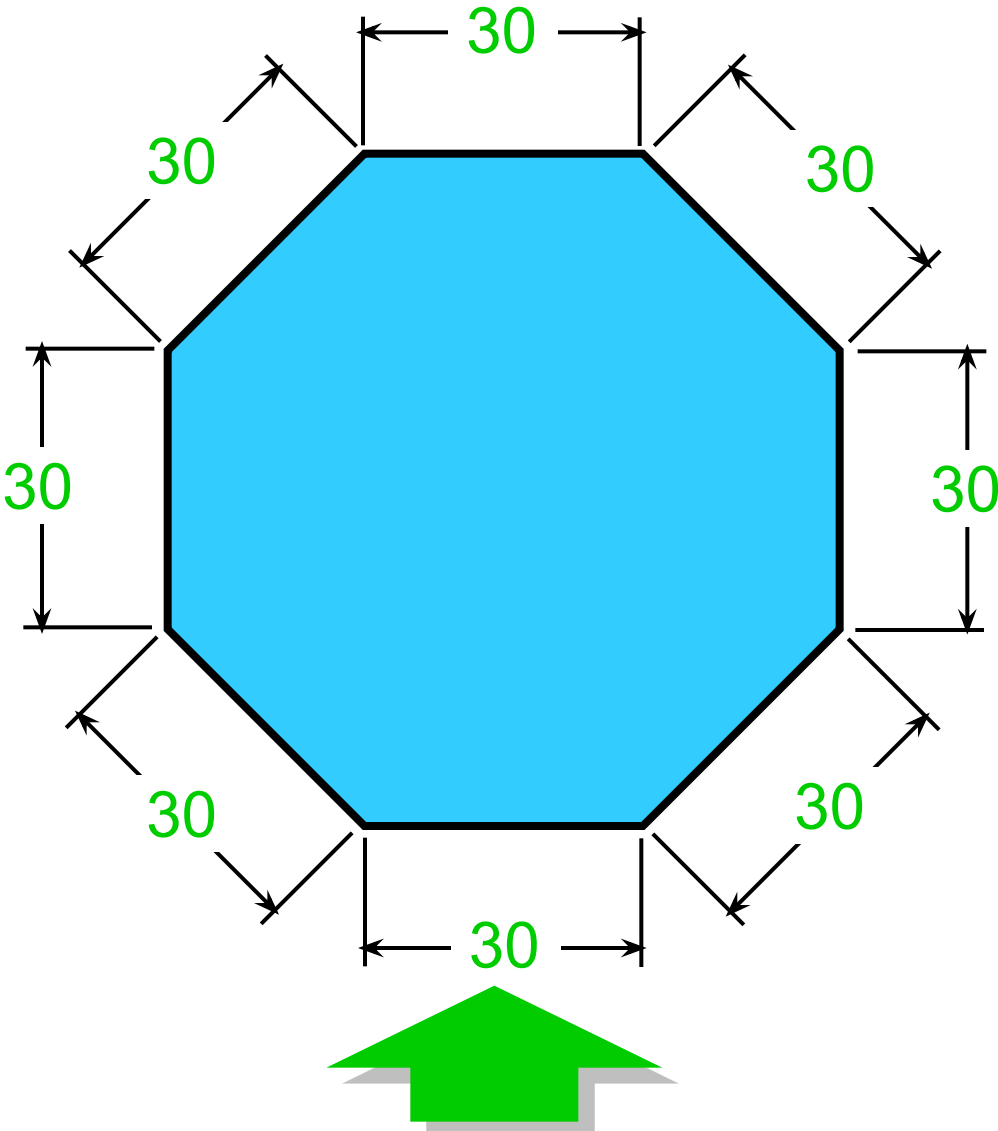
The dimension figures are placed so that they can be read from the **bottom** of the drawing.

Do not use both system on the same drawing or on the same series of drawing

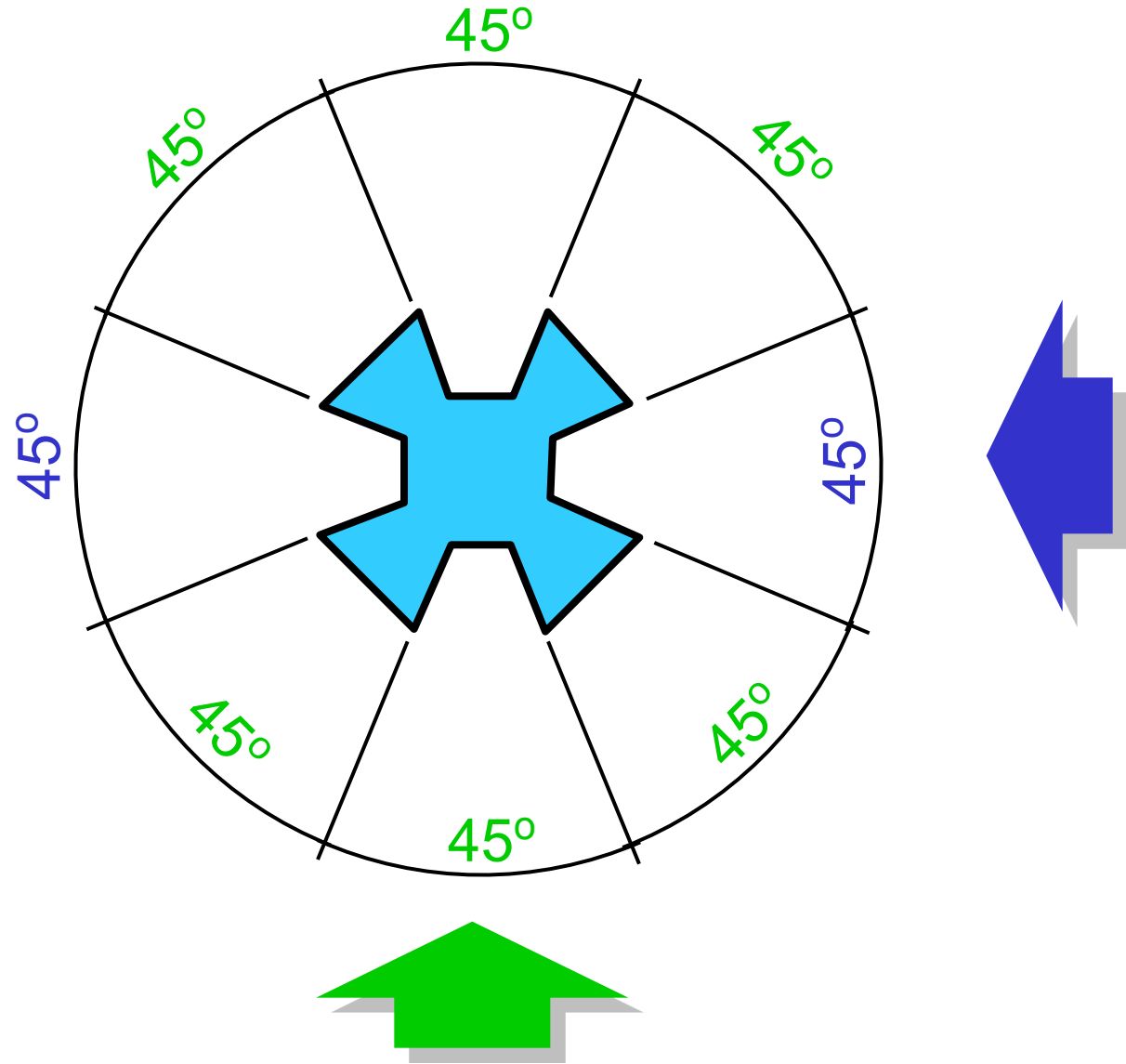
EXAMPLE : Dimension of *length* using *aligned* method.



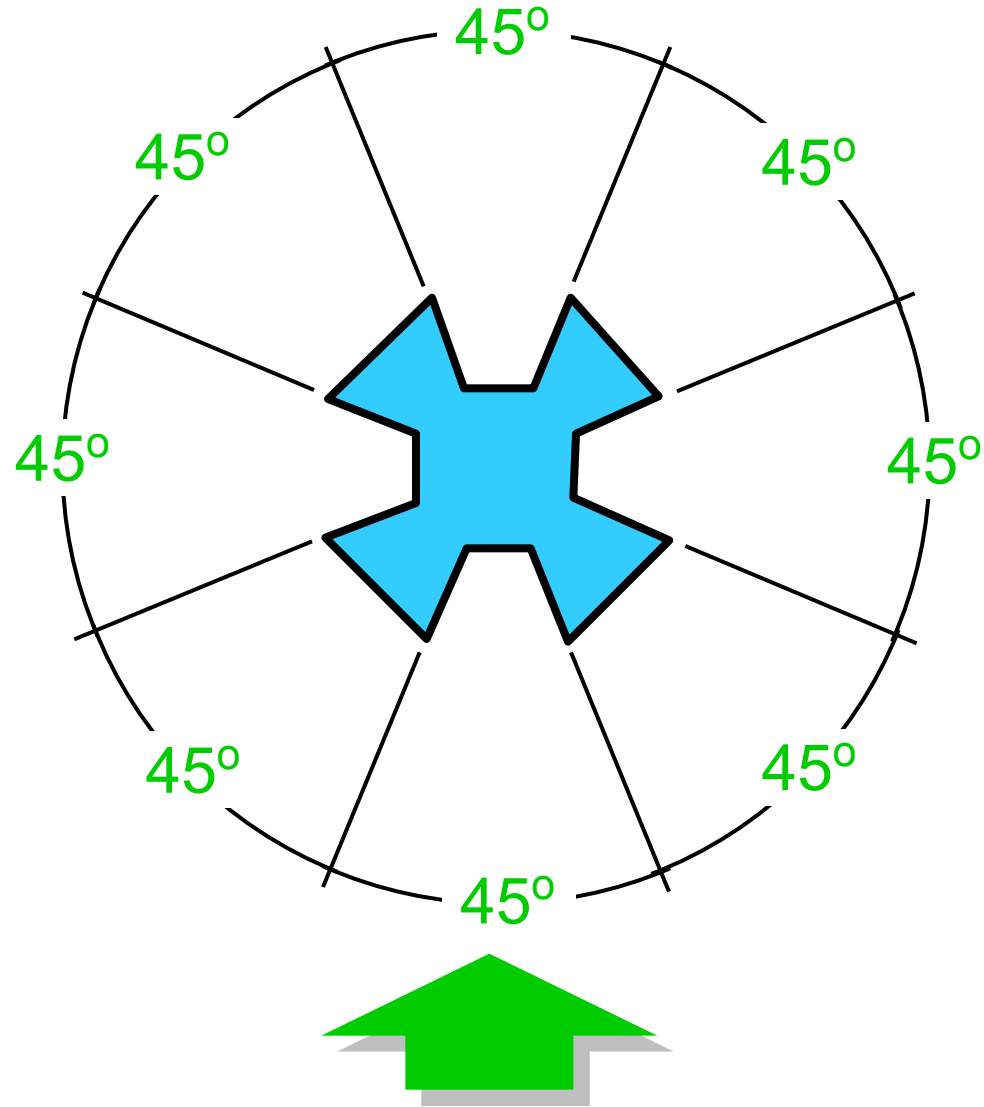
EXAMPLE : Dimension of *length* using *unidirectional* method.



EXAMPLE : Dimension of *angle* using *aligned* method.

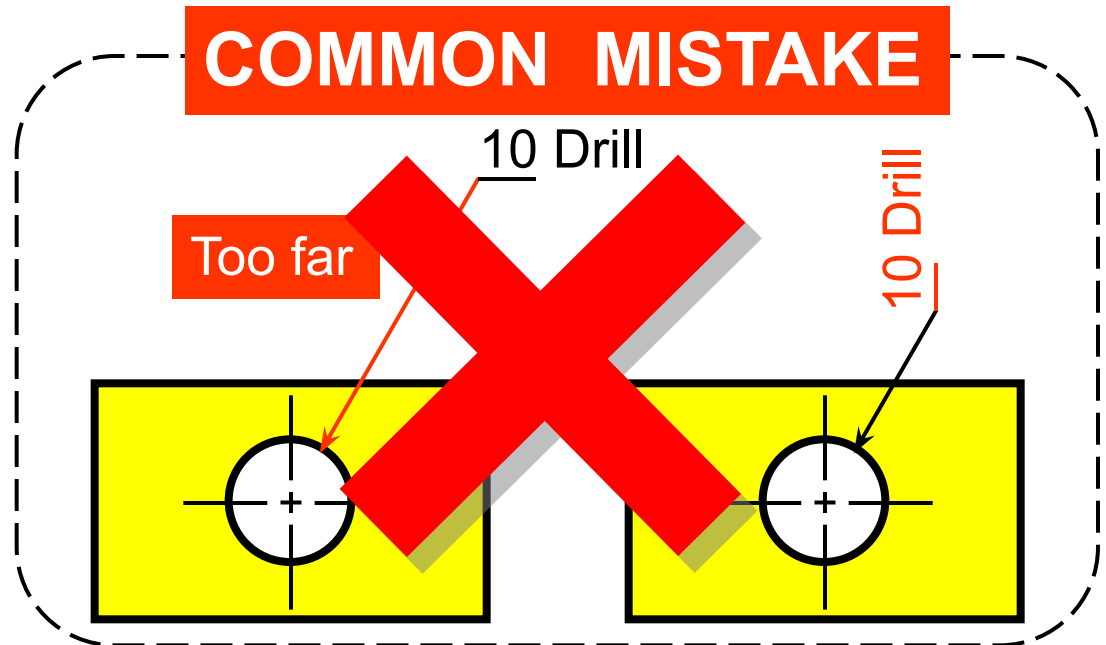
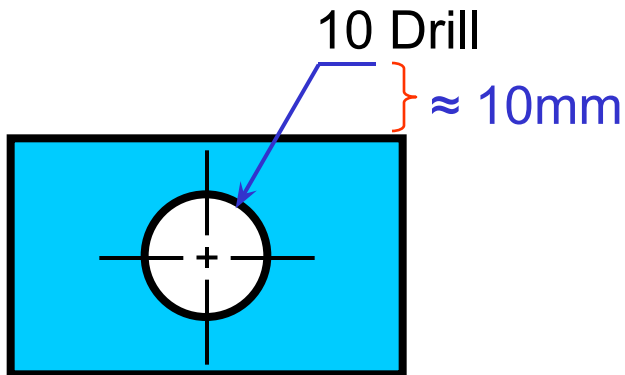


EXAMPLE : Dimension of *angle* using *unidirectional* method.



LOCAL NOTES

- Place the notes **near** to the feature which they apply, and should be placed outside the view.
- Always read **horizontally**.



Dimensioning Practices



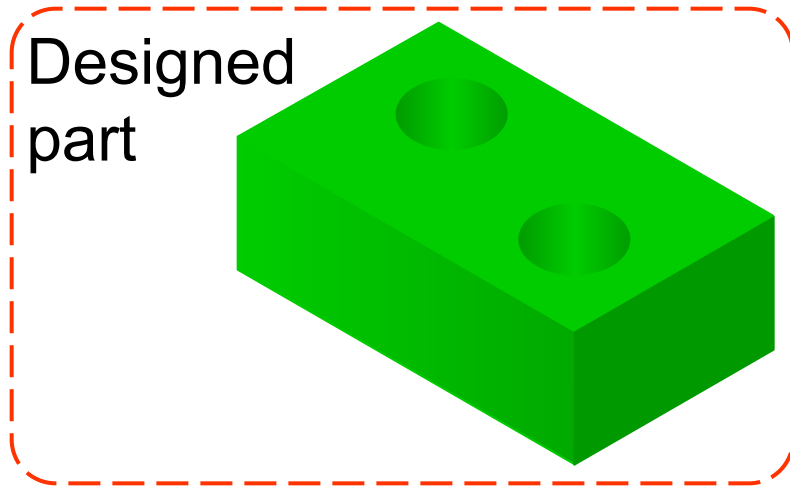
THE BASIC CONCEPT

Dimensioning is accomplished by adding *size* and *location* information *necessary to manufacture* the object.

This information have to be

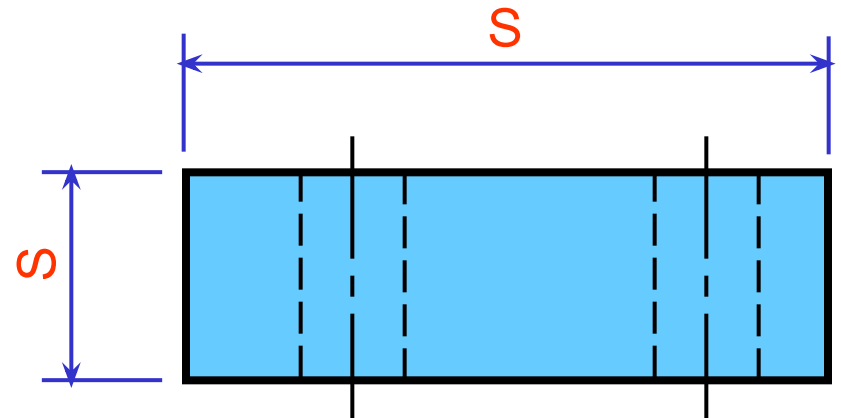
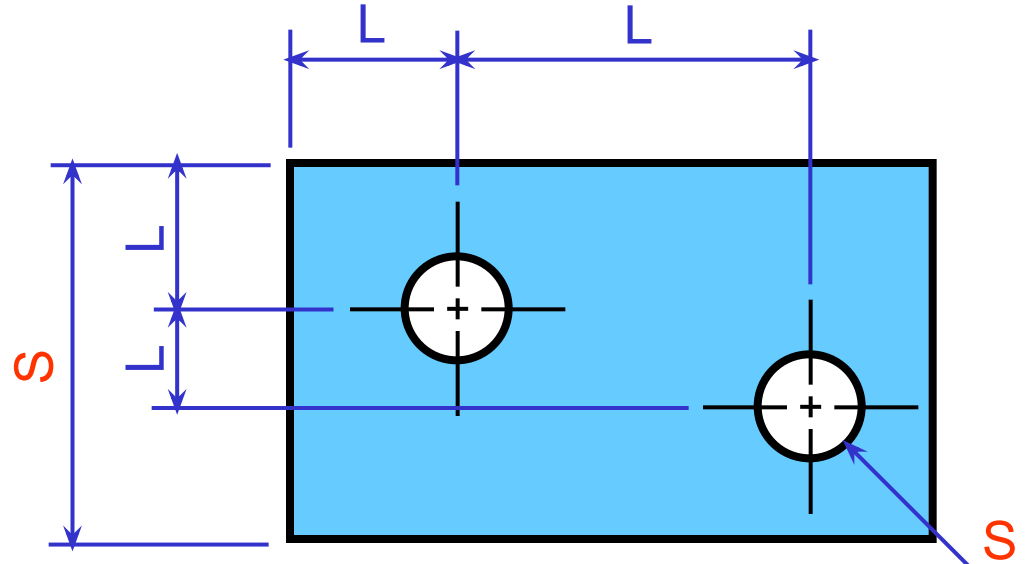
- Clear
- Complete
- Facilitate the
 - manufacturing method
 - measurement method

EXAMPLE



To manufacture this part we need to know...

1. **Width**, **depth** and **thickness** of the part.
2. **Diameter** and **depth** of the hole.
3. **Location** of the holes.

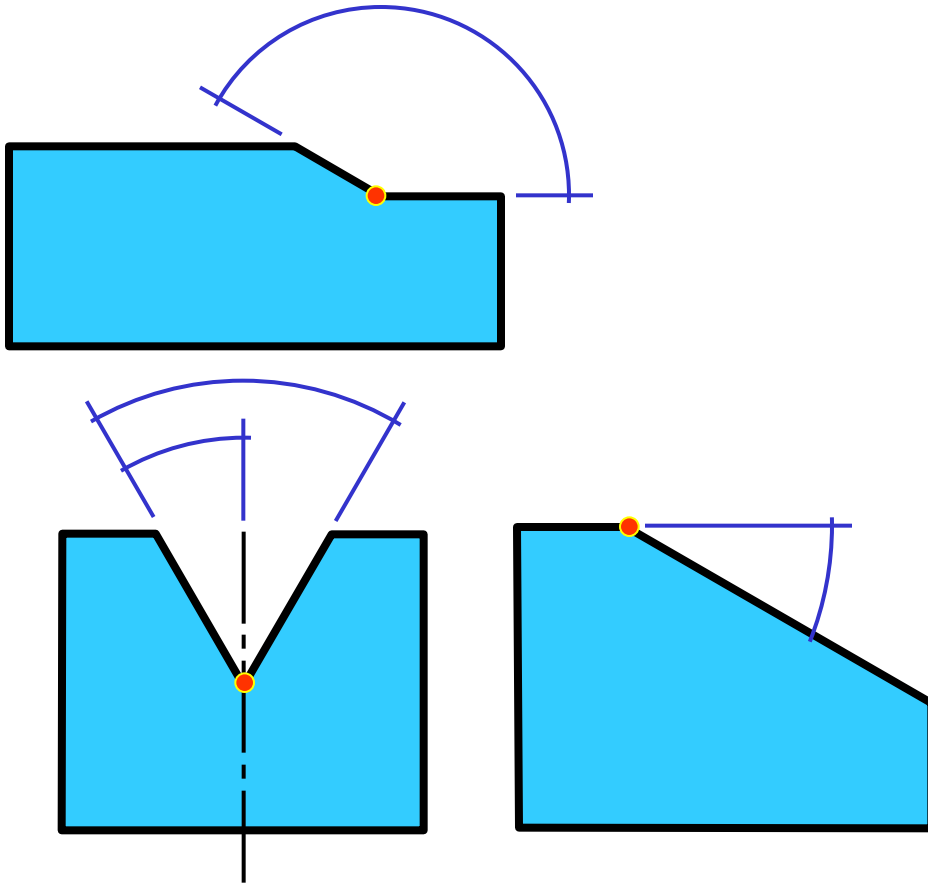


“S” denotes size dimension.

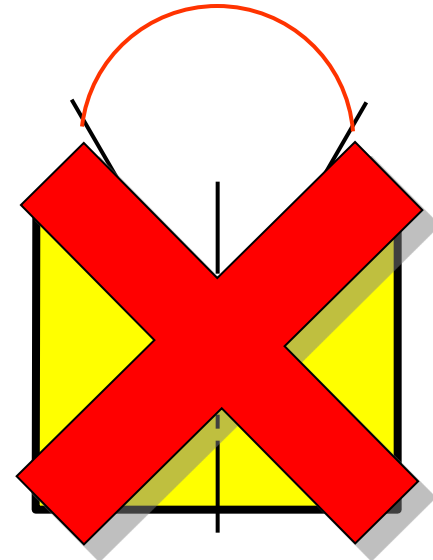
“L” denotes location dimension.

ANGLE

- To dimension an angle use **circular dimension line** having the center at the vertex of the angle.

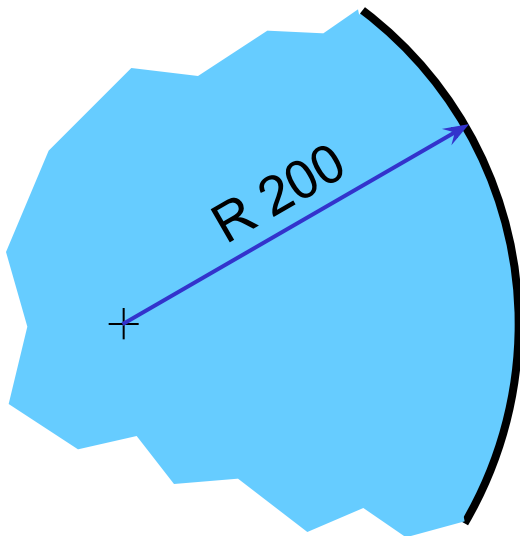


COMMON MISTAKE

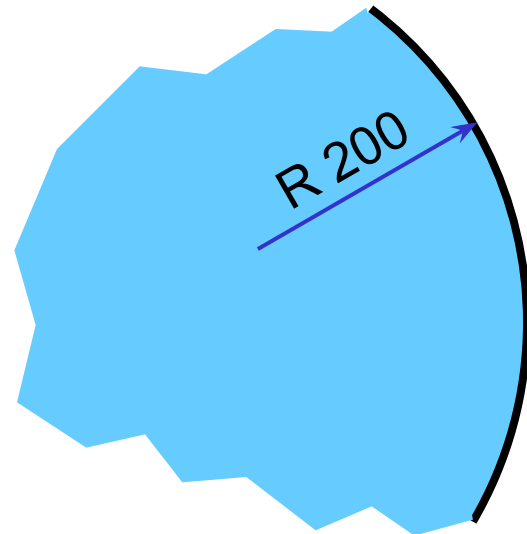


ARC

- Arcs are dimensioned by giving the **radius**, in the views in which their true shapes appear.
- The letter “R” is **always** lettered before the figures to emphasize that this dimension is radius of an arc.



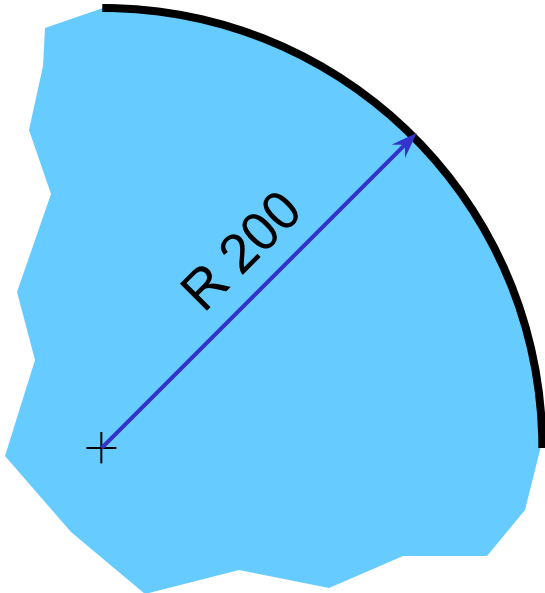
or



ARC

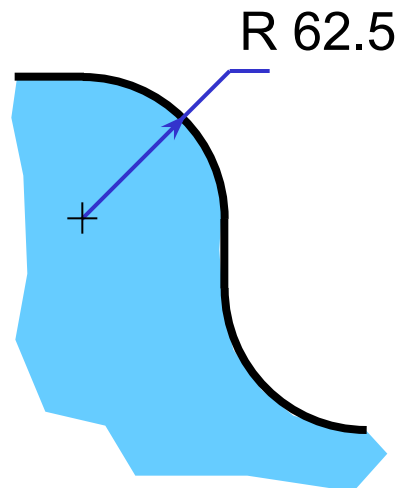
- The dimension figure and the arrowhead **should be inside** the arc, where there is sufficient space.

Sufficient space
for both.



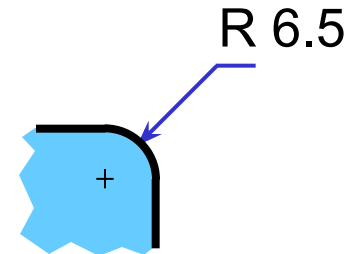
Sufficient space
for arrowhead only.

Move figure outside



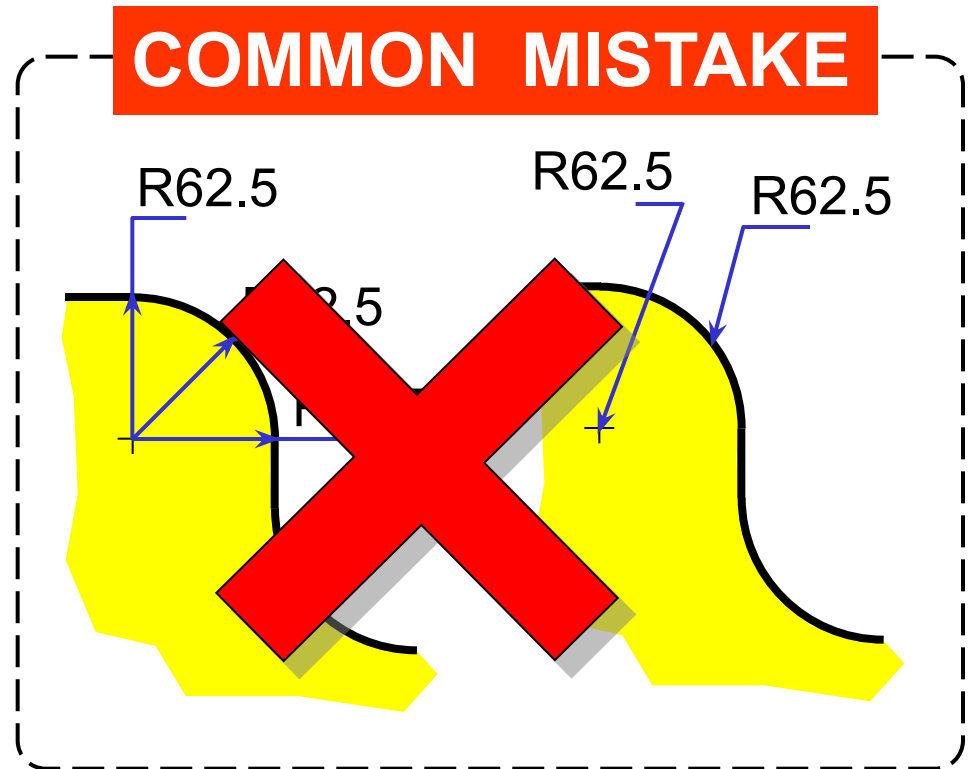
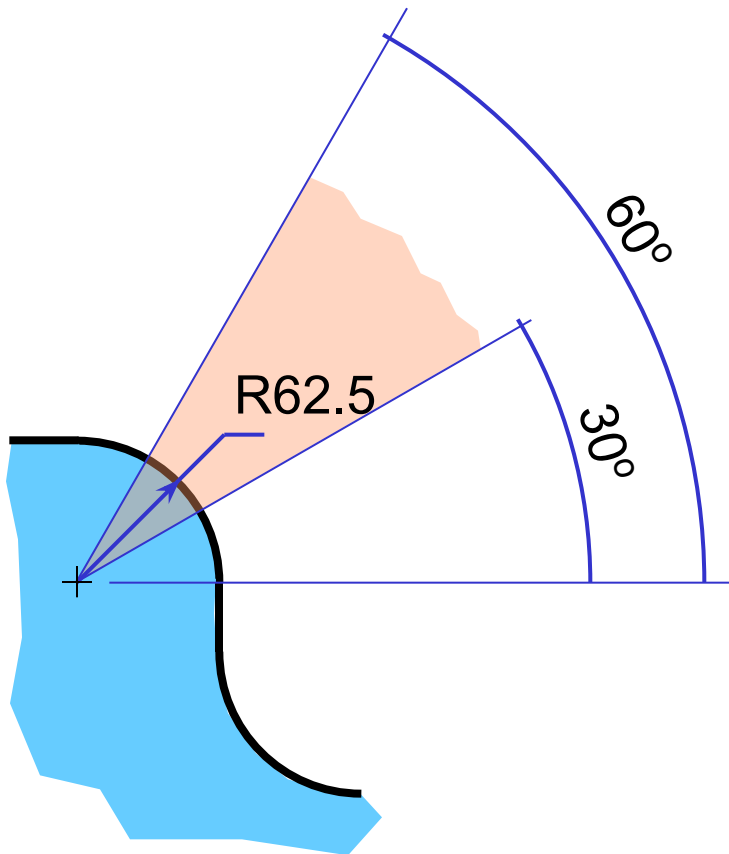
Insufficient space
for both.

Move both figure
and arrow outside



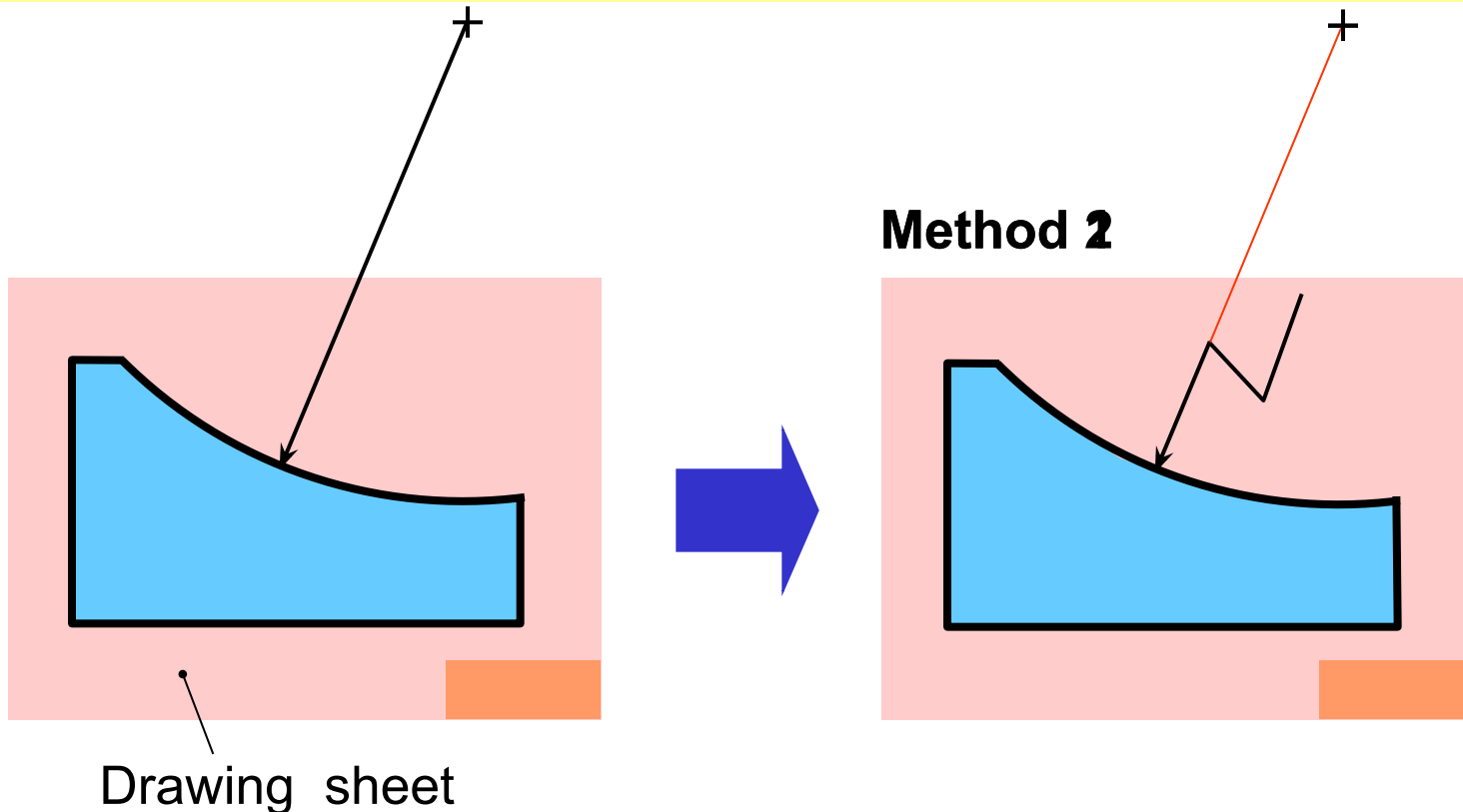
ARC

- Leader line **must** be **radial** and **inclined** with an angle between 30 ~ 60 degs to the horizontal.



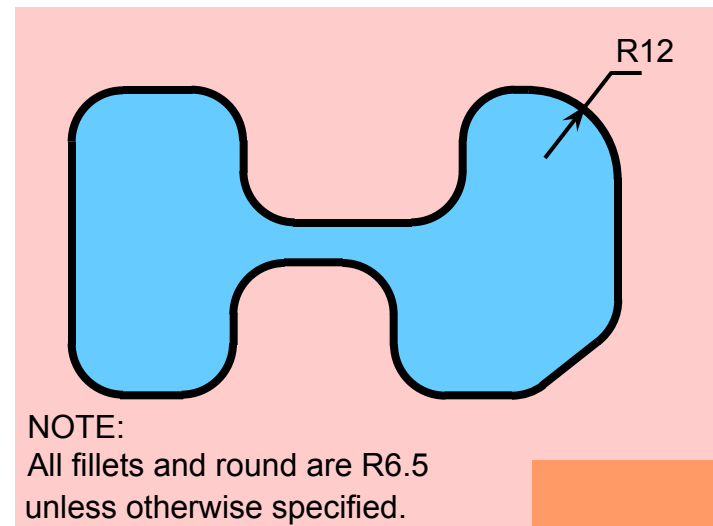
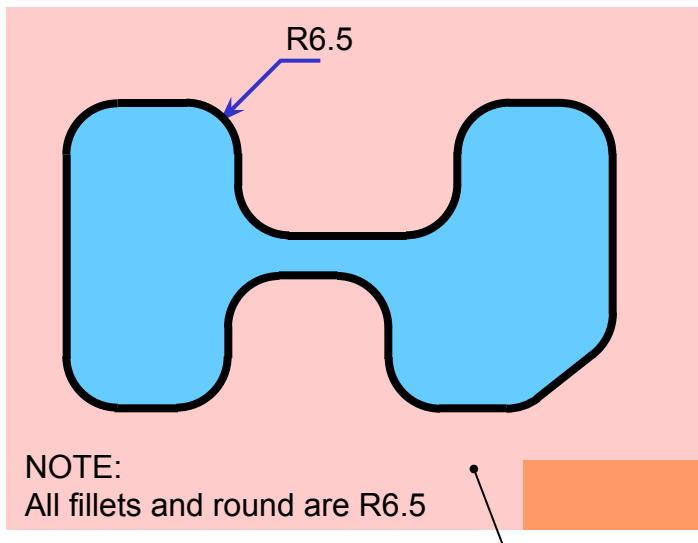
ARC

- Use the **foreshortened radial dimension line**, when arc's center locates outside the sheet or interfere with other views.



FILLETS AND ROUNDS

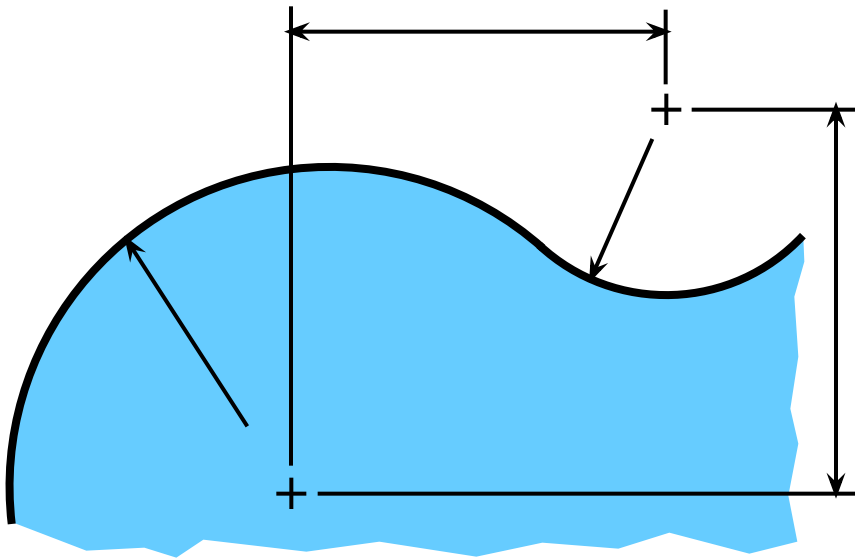
- Give the radius of a typical fillet only by using a *local* note.
- If all fillets and rounds are uniform in size, dimension may be omitted, but it is necessary to add the note “ All fillets and round are Rxx. ”



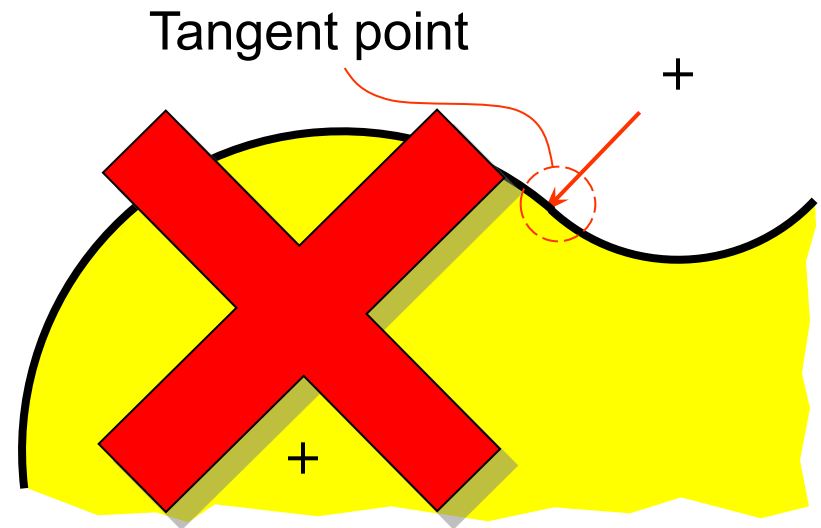
Drawing sheet

CURVE

- The curve constructed from two or more arcs, requires the dimensions of **radii** and **center's location**.



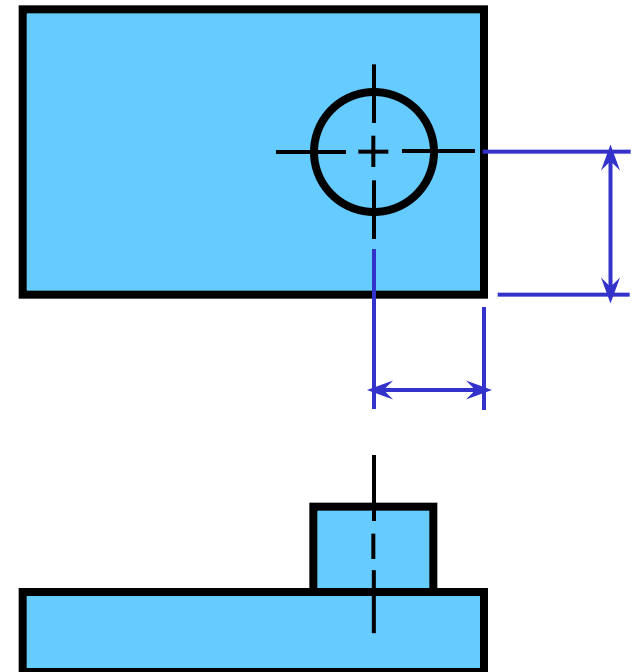
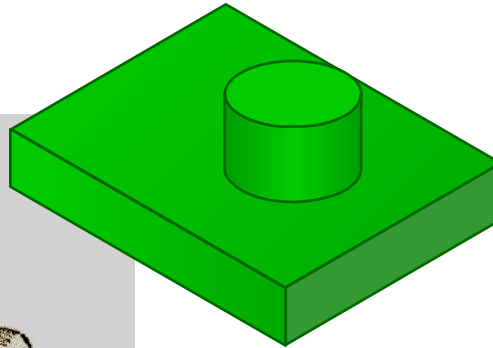
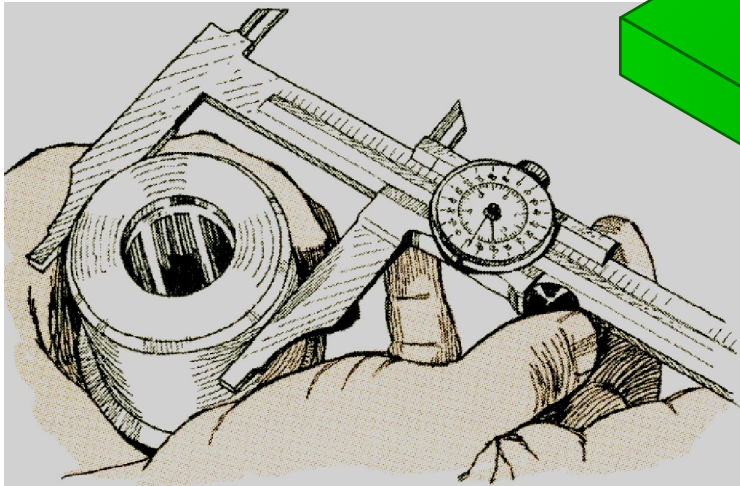
COMMON MISTAKE



CYLINDER

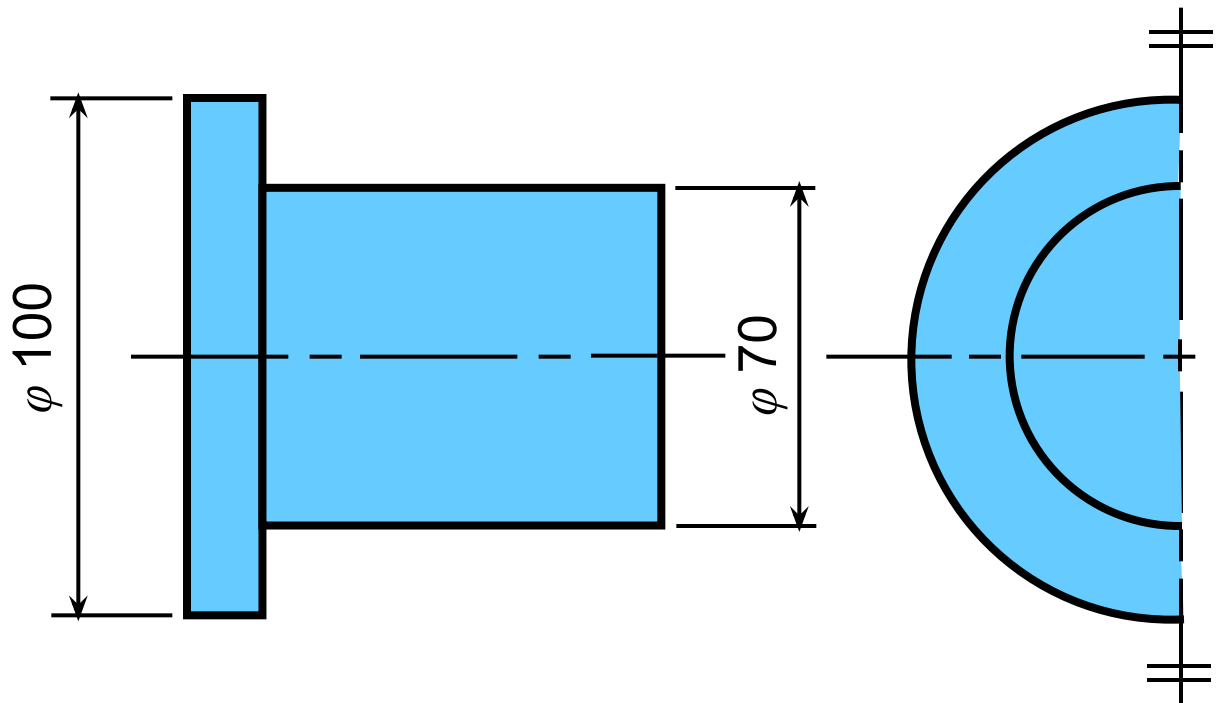
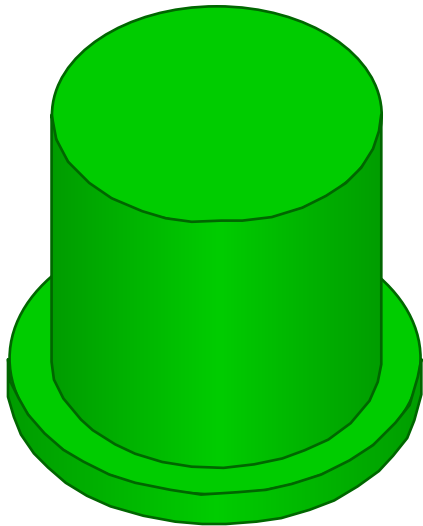
- Size dimensions are **diameter** and **length**.
- Location dimension **must** be located from its center lines and **should be** given in circular view.

Measurement method



CYLINDER

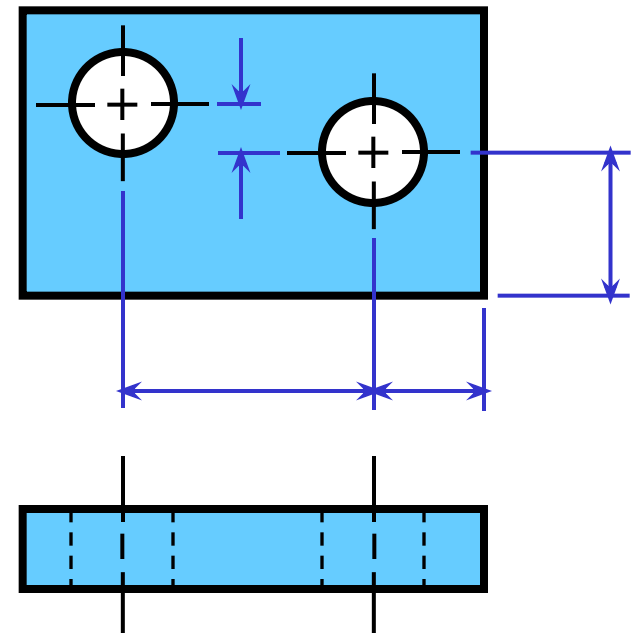
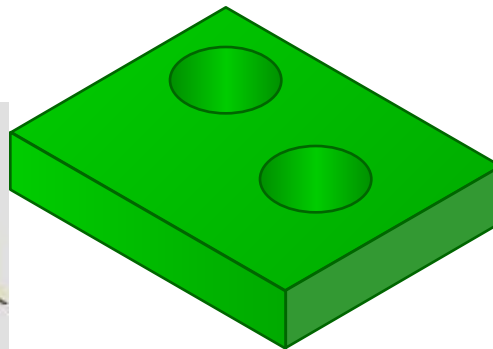
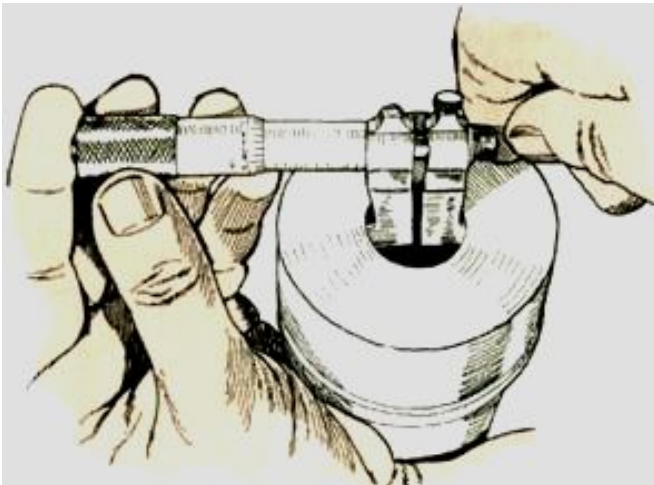
- Diameter **should be** given in a **longitudinal view** with the symbol “ ϕ ” placed before the figures.



HOLES

- Size dimensions are **diameter** and **depth**.
- Location dimension **must** be located from its center lines and should be given in circular view.

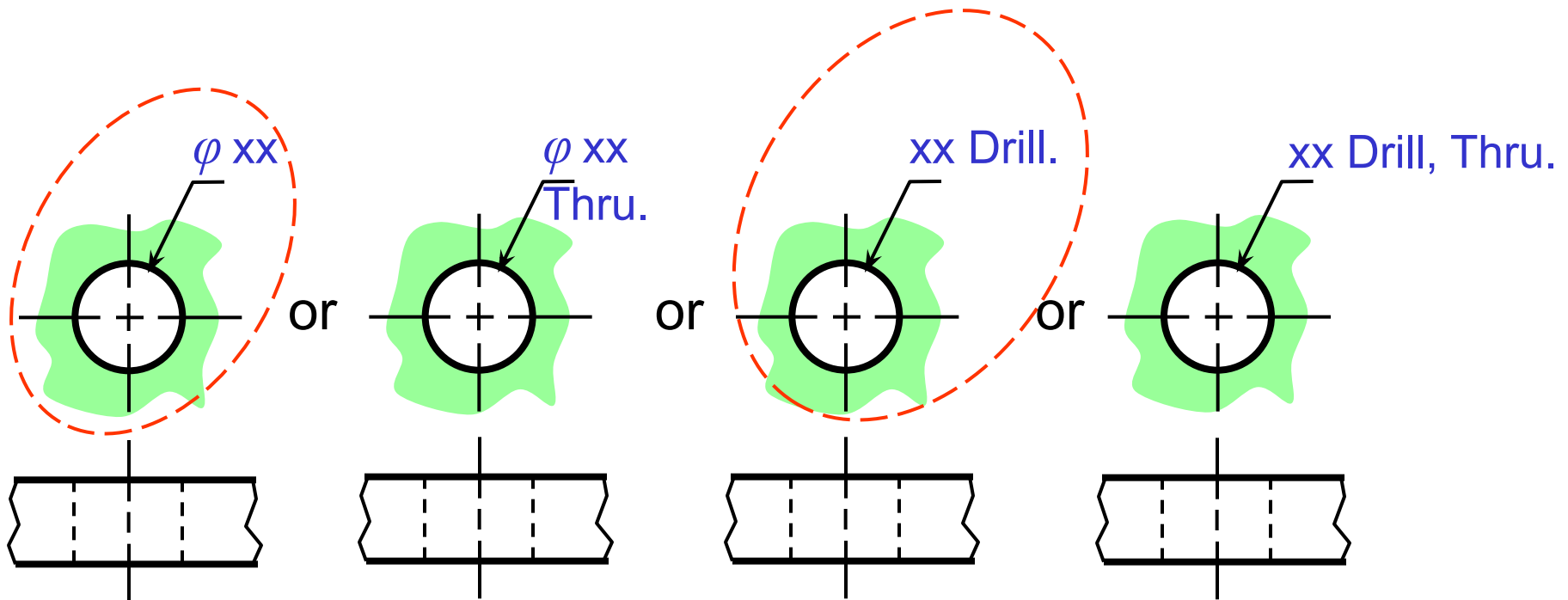
Measurement method



HOLES : SMALL SIZE

- Use leader line and local note to specify **diameter** and **hole's depth** in the circular view.

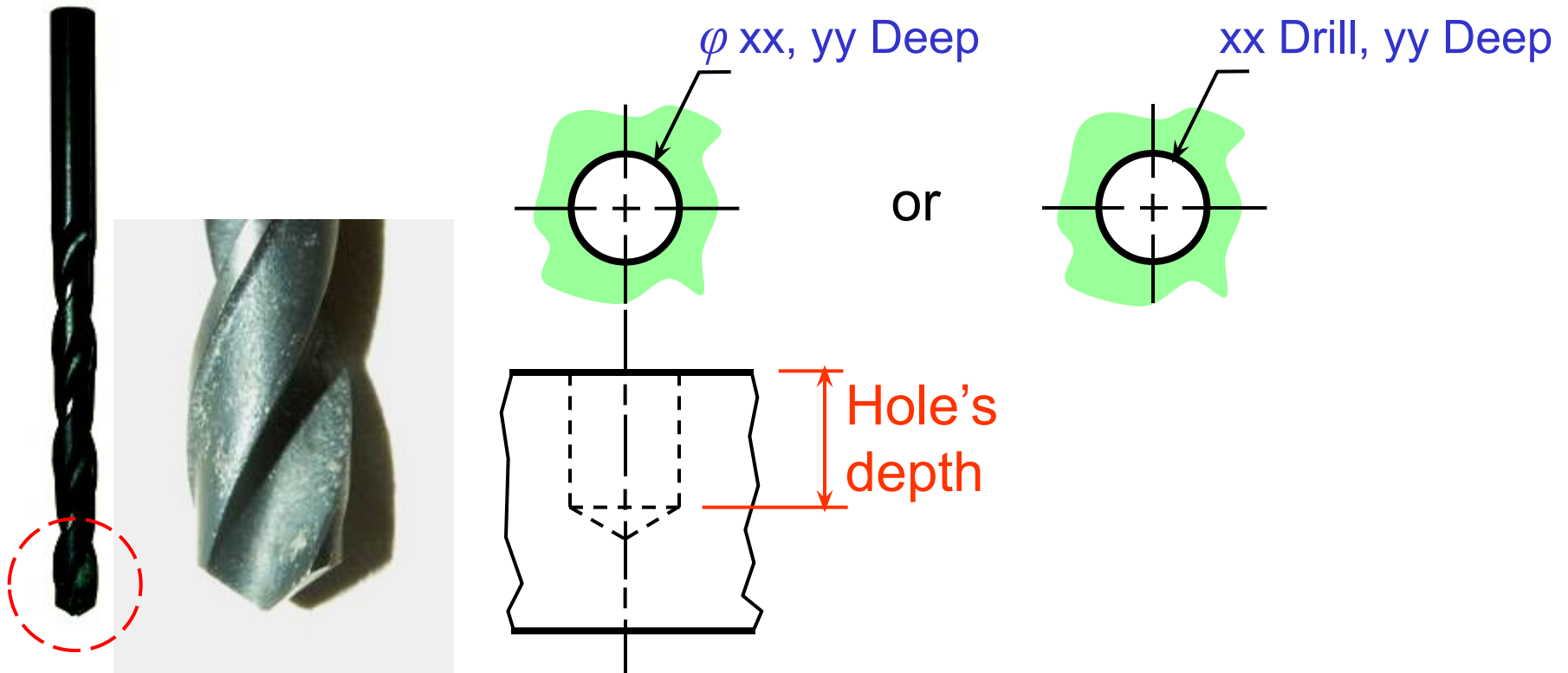
1) Through thickness hole



HOLES : SMALL SIZE

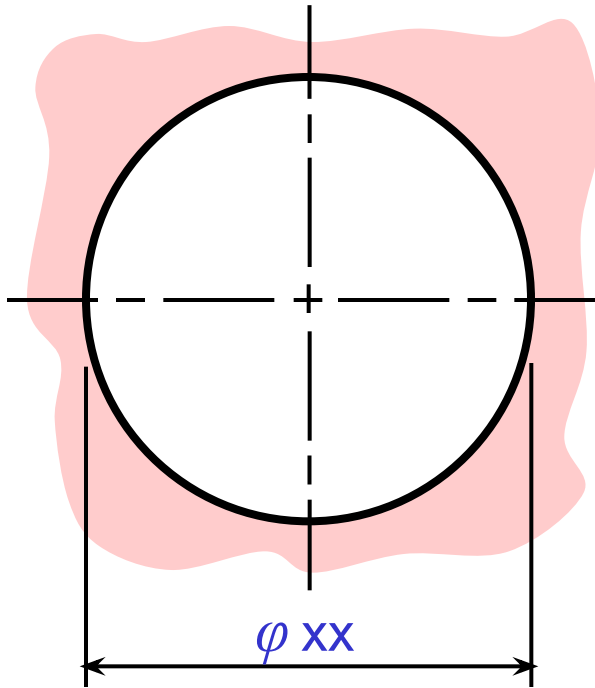
- Use leader line and local note to specify **diameter** and **hole's depth** in the circular view.

2) Blind hole

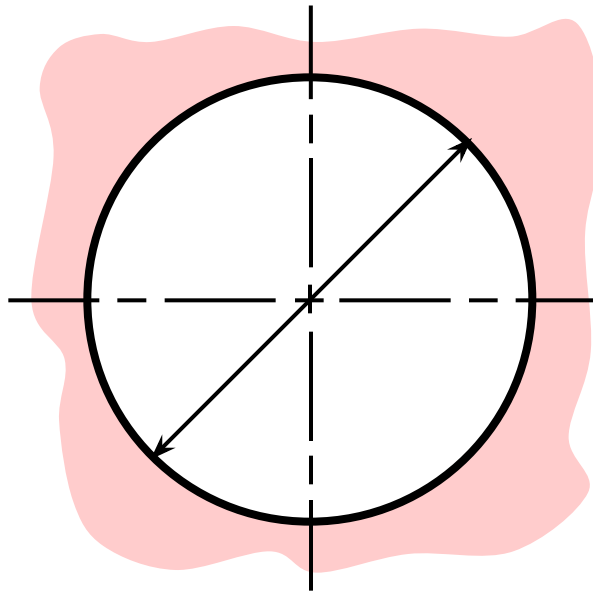


HOLES : LARGE SIZE

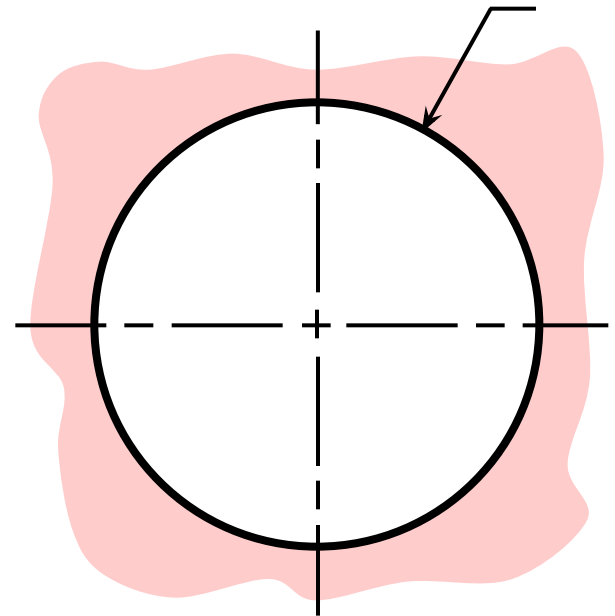
Use extension and
dimension lines



Use diametral
dimension line

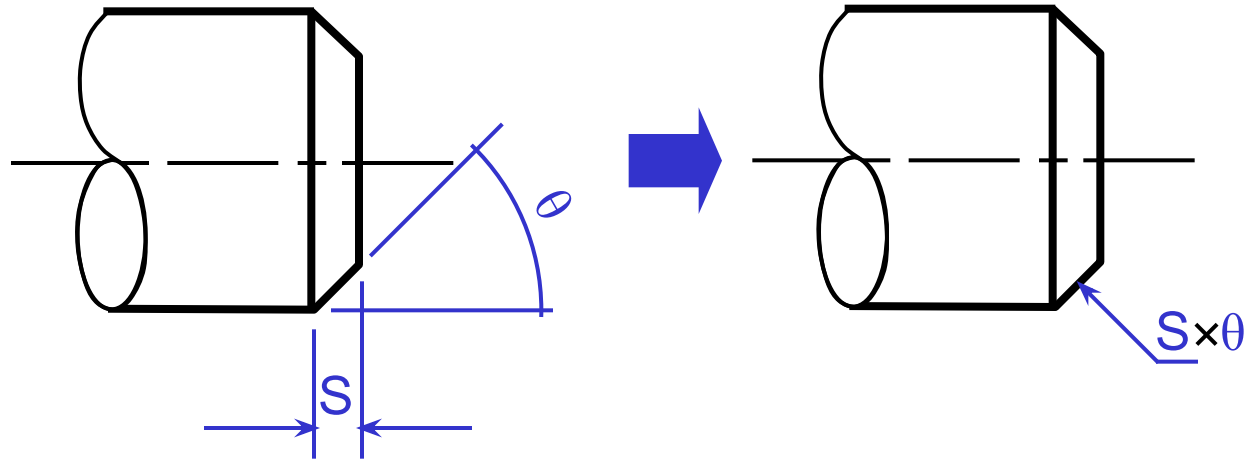
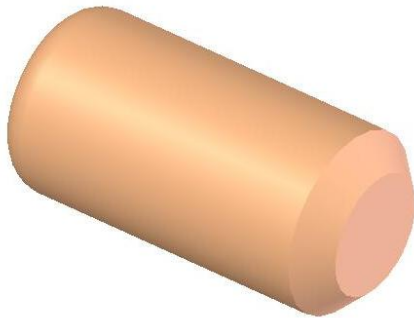


Use leader line
and note

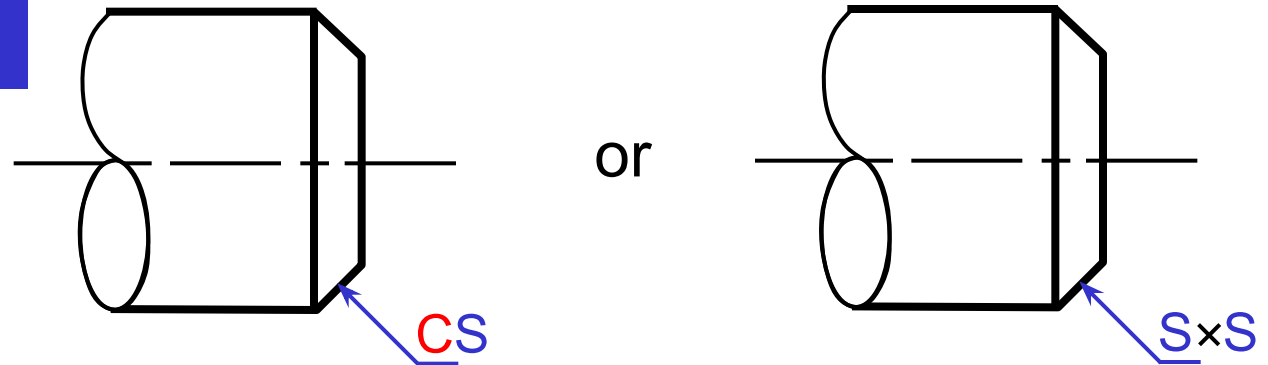


CHAMFER

- Use leader line and note to indicate **linear distance** and **angle** of the chamfer.



For a 45° chamfer



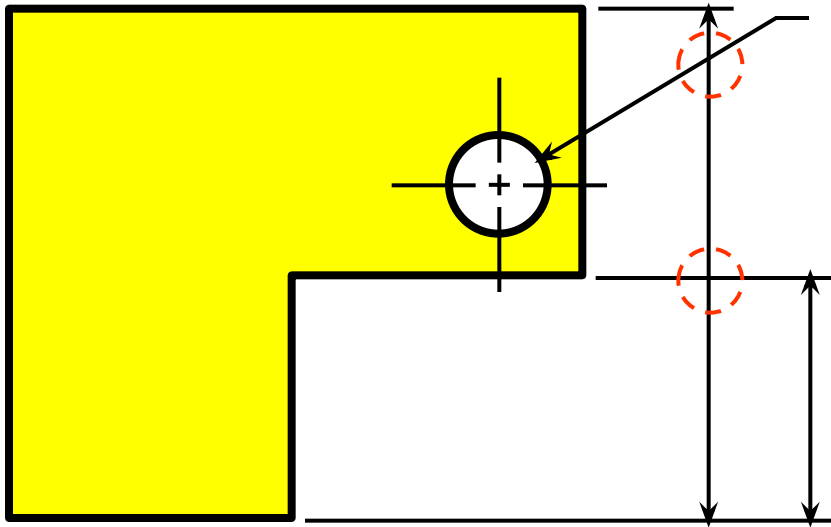
Placement of Dimensions



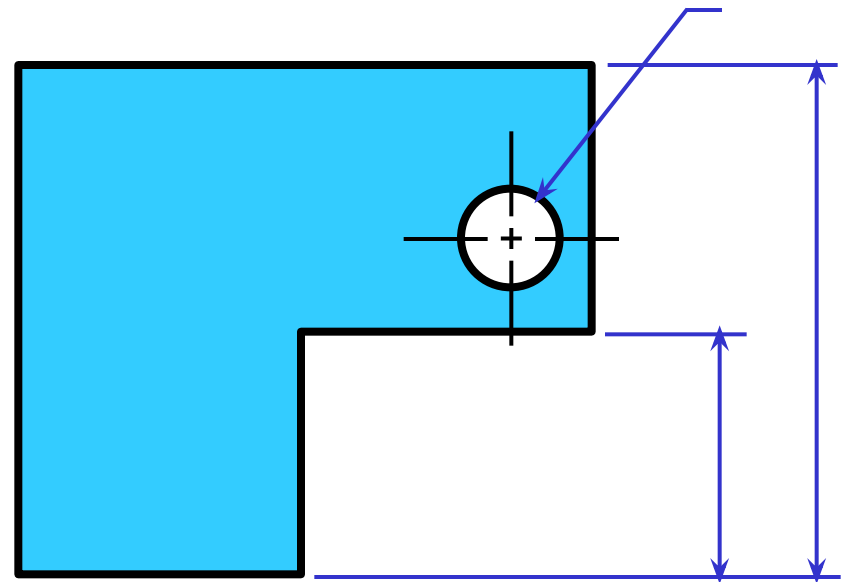
RECOMMENDED PRACTICE

1. Extension lines, leader lines **should not** cross dimension lines.

POOR



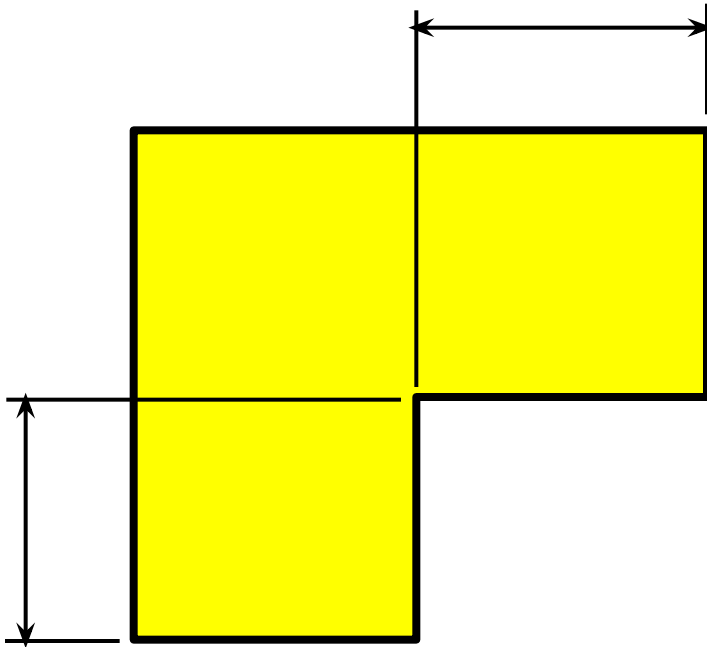
GOOD



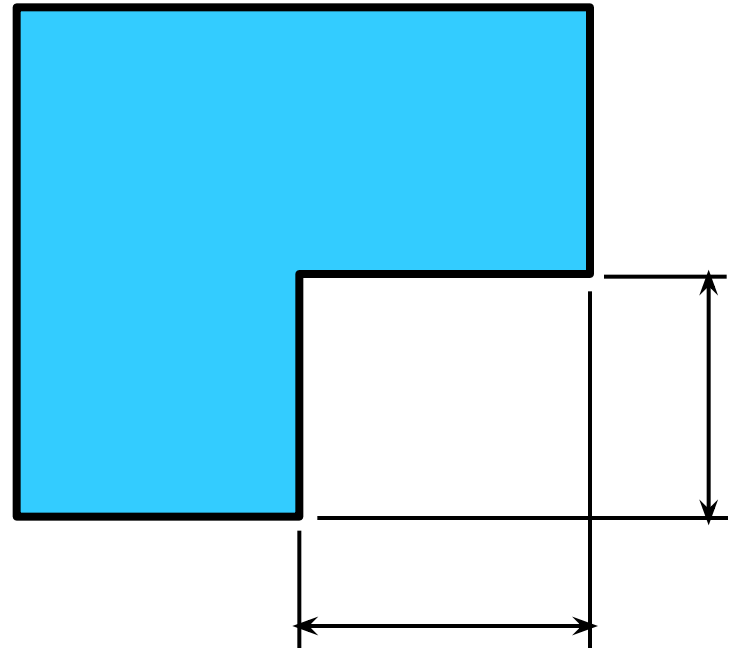
RECOMMENDED PRACTICE

2. Extension lines **should be** drawn from the nearest points to be dimensioned.

POOR



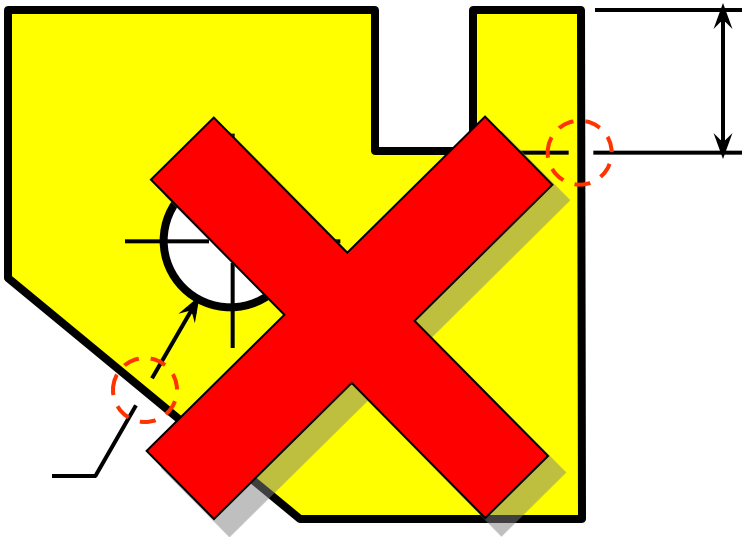
GOOD



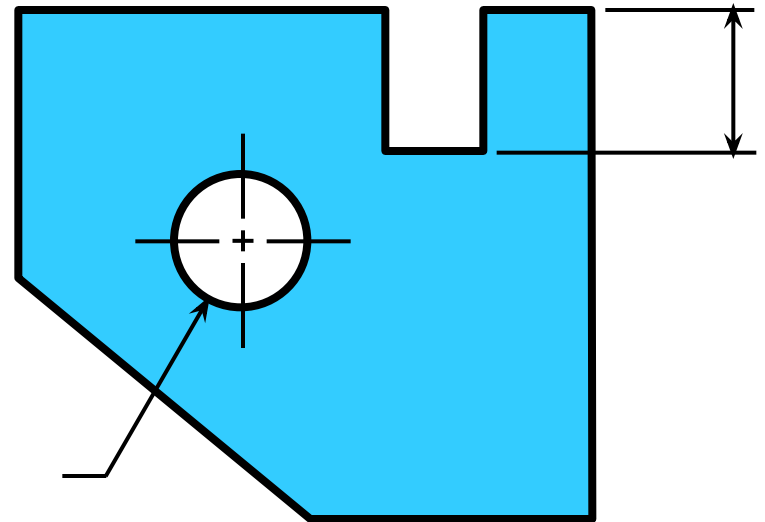
RECOMMENDED PRACTICE

3. *Extension lines of internal feature **can** cross visible lines **without** leaving a gap at the intersection point.*

WRONG



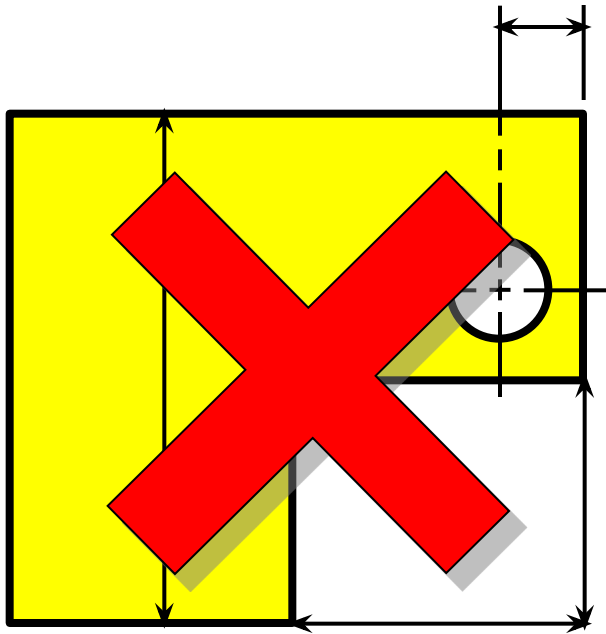
CORRECT



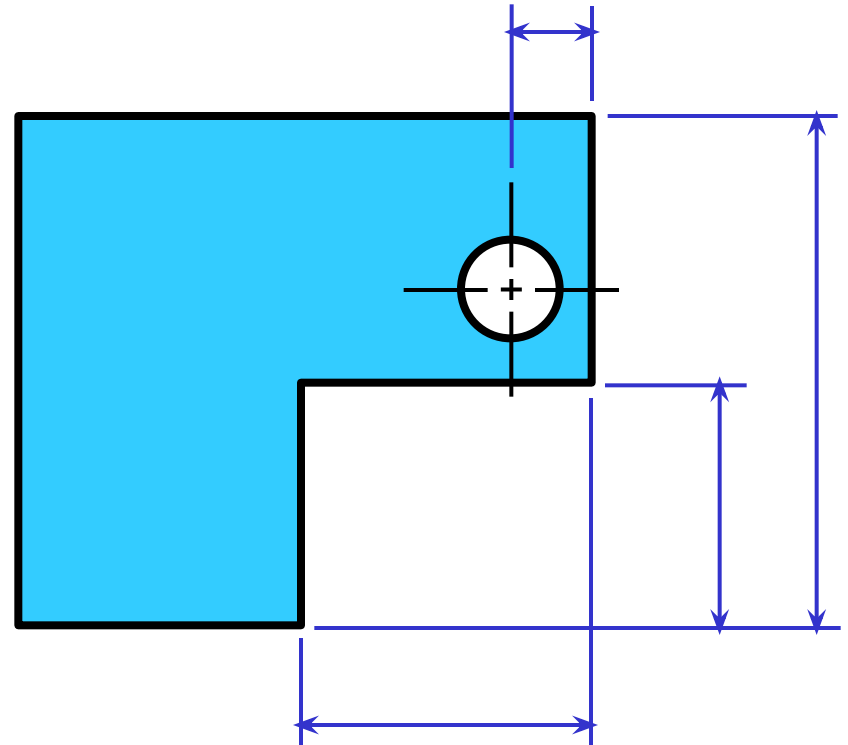
RECOMMENDED PRACTICE

4. **Do not** use *object line*, *center line*, and *dimension line* as an extension lines.

POOR



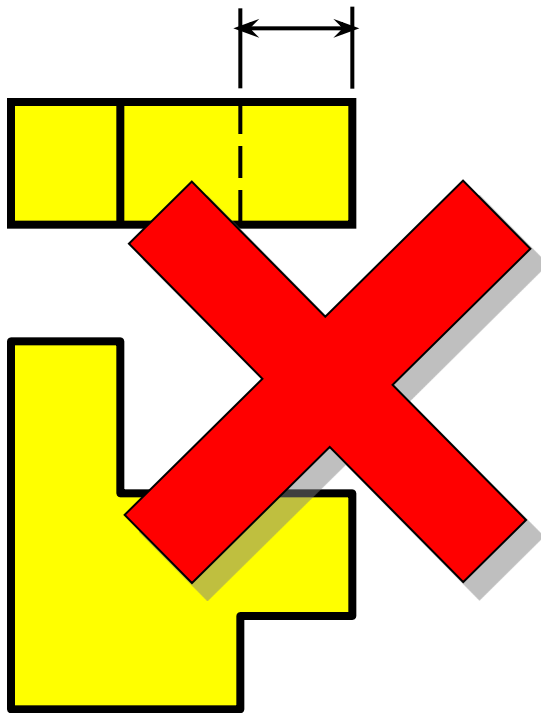
GOOD



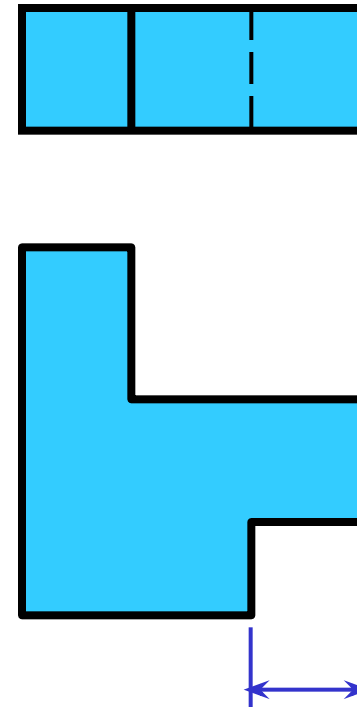
RECOMMENDED PRACTICE

5. ***Avoid** dimensioning hidden lines.*

POOR



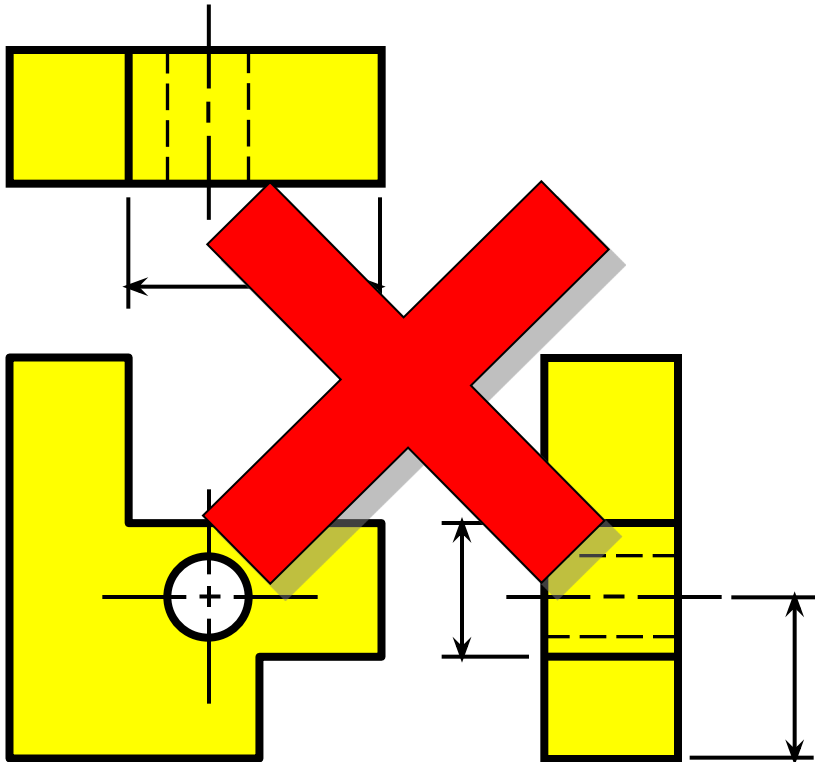
GOOD



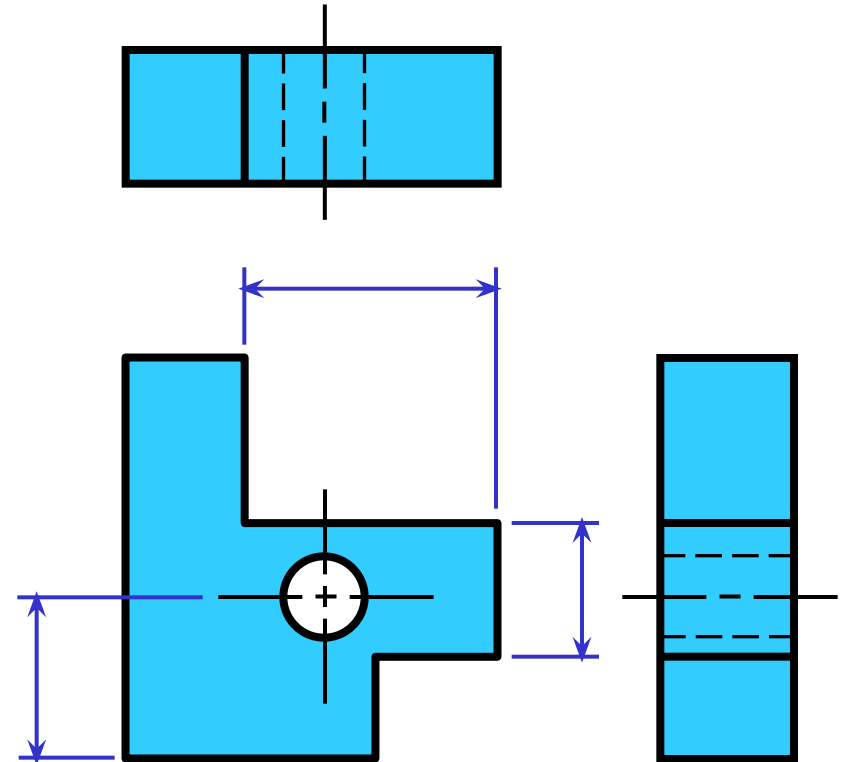
RECOMMENDED PRACTICE

6. Apply the dimension to the view that clearly show the shape or features of an object.

POOR



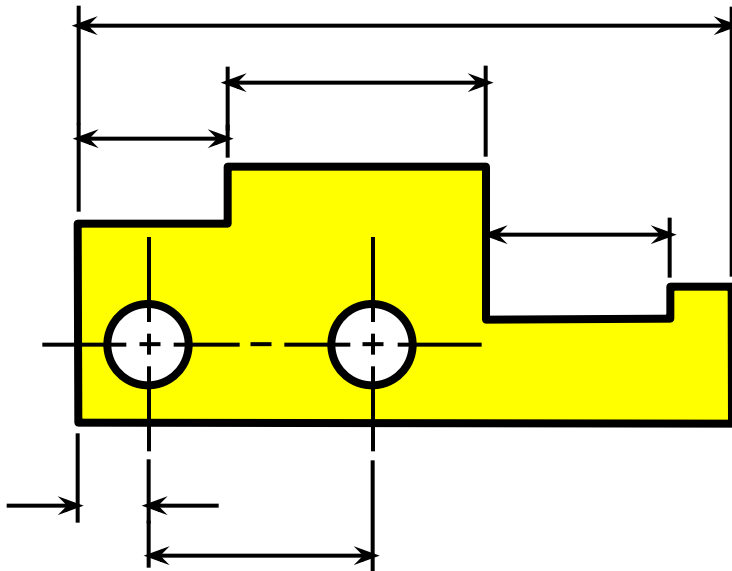
GOOD



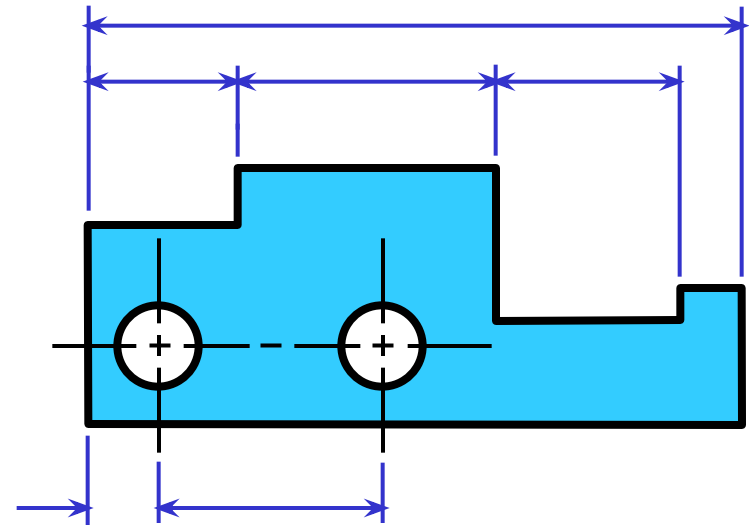
RECOMMENDED PRACTICE

7. Dimension lines should be lined up and grouped together as much as possible.

POOR



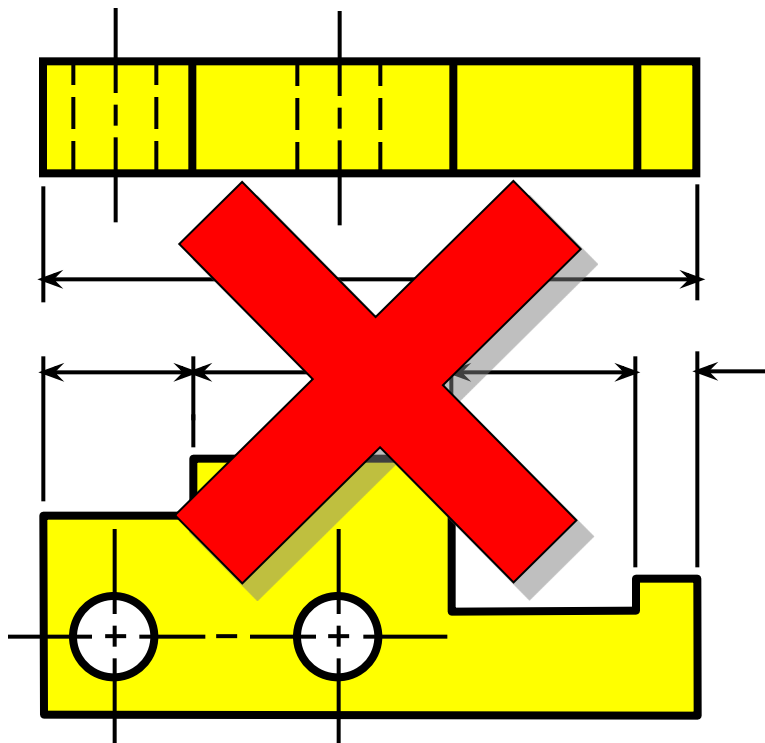
GOOD



RECOMMENDED PRACTICE

8. *Do not repeat a dimension.*

POOR



GOOD

